Stochastic Financial Models

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Financial markets are, from a mathematician's point of view, a rather remarkable object of study. They are made of people, and people are not easy to axiomatise. And yet if we grant ourselves a single principle – that one cannot make something out of nothing – then a startling amount of structure falls out. Prices of complicated derivative securities are pinned down exactly by the prices of simple ones; the 'right' probability measure to compute with turns out not to be the one describing what will actually happen; and the whole edifice rests on the theory of martingales.

This course is really two courses braided together. One strand is economic: how should an agent with preferences choose a portfolio, and what does market equilibrium look like? The other strand is probabilistic: conditional expectation, martingales, optional stopping, Brownian motion, and a first look at stochastic calculus. The two strands meet in the *fundamental theorem of asset pricing*, which says that a market admits no arbitrage precisely when there is an equivalent measure making discounted prices into martingales. Everything after that theorem is, in one way or another, a computation of a conditional expectation.

This article constitutes my notes for the 'Stochastic Financial Models' course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

Contents

§1 The One-Period Market

Before we can prove anything, we need a model. The models in this course are deliberately, almost offensively, simple; the justification is that even the simplest of them already produces the Black–Scholes formula, and that the qualitative conclusions turn out to be robust.

§1.1 Standing Assumptions

We shall work throughout under the following idealisations, which we state once and then quietly forget.

(i) *No transaction costs.* Buying and selling are free.

(ii) *Infinite divisibility.* We may hold any real number of units of any asset, not just an integer number.

(iii) *No bid–ask spread.* There is one price at which an asset may be both bought and sold.

(iv) *No price impact* ('linear pricing'). Buying 2θ units costs exactly twice as much as buying θ units, however large θ is. Our trading does not move the market.

(v) *No short-selling constraints.* We may hold a negative number of units of an asset.

(vi) *Borrowing and lending at the same rate.* There is a riskless bank account paying interest at rate r, and we may deposit into it or borrow from it freely.

Assumption (iv) is the one doing the most work, and it is also the one most obviously false: if you try to buy every share in a company you will find the price moves against you. But without it there is no hope of a linear theory, and linearity is what makes everything computable.

§1.2 Portfolios and Wealth

Our basic market has d **risky assets**, whose prices at time t we write as a random vector

$$S_t = (S_t^1, \dots, S_t^d)^\top.$$

For the moment there are only two times, $t = 0$ and $t = 1$; this is the **one-period model**. The price S_0 is known at time 0, and S_1 is a random vector whose distribution we assume known. In addition there is a **riskless asset** (asset 0, the bank account) which turns one pound at time 0 into $1 + r$ pounds at time 1.

Definition 1.1 (Portfolio)
A **portfolio** is a vector $\theta = (\theta^1, \dots, \theta^d)^\top \in \mathbb{R}^d$, where θ^i is the number of units of asset i held between time 0 and time 1, together with a number $\theta^0 \in \mathbb{R}$ of pounds in the bank.

If $\theta^i > 0$ we say the agent is **long** asset i; if $\theta^i < 0$ the agent is **short** it, meaning that they have borrowed the asset and sold it, and must return it at time 1.

Let X_t denote the agent's wealth at time t, with $X_0 = x$ given. Since the portfolio must be paid for out of the initial wealth, we have the **budget constraint**

$$X_0 = \theta^0 + \theta^\top S_0.$$

At time 1 the bank deposit has grown and the shares are worth $\theta^\top S_1$, so

$$X_1 = \theta^0(1+r) + \theta^\top S_1 = (1+r)\left(X_0 - \theta^\top S_0\right) + \theta^\top S_1.$$

Rearranging gives an expression which will be with us for the rest of the course:

$$\boxed{X_1 = (1+r)X_0 + \theta^\top\left(S_1 - (1+r)S_0\right).}$$

In other words – the wealth at time 1 is what you would have got by leaving everything in the bank, plus a term measuring how much better the risky assets did than the bank. We give that term a name.

Definition 1.2 (Excess return)
The **excess return vector** of the market is

$$\xi = S_1 - (1+r)S_0 \in \mathbb{R}^d.$$

So $X_1 = (1+r)X_0 + \theta^\top\xi$, and the whole of the one-period theory is really the study of the random vector ξ. Note that ξ is genuinely random – it is the only randomness in the model – and that the map $\theta \mapsto \theta^\top\xi$ is linear. Almost every theorem below is a statement about the linear subspace $\{\theta^\top\xi : \theta \in \mathbb{R}^d\}$ of random variables.

We write $\mathcal{X}(x) = \{(1+r)x + \theta^\top\xi : \theta \in \mathbb{R}^d\}$ for the set of **attainable** time-1 wealths starting from initial wealth x. This is an affine subspace of the space of random variables.

Remark. It is worth pausing to note what has *not* been assumed. We have not assumed that the agent knows anything about S_1 beyond its distribution, and we have certainly not assumed that this distribution is correct. Estimating it is a large industry (and something you might recognise from Part IB Statistics); it is not our concern here.

§1.3 A First Example

Let us immediately do a computation, so that the abstractions have something to hang on.

Example 1.3 (A one-period binomial market)
Take $d = 1$, interest rate $r = 0$, and $S_0 = 5$. Suppose S_1 takes the value 7 or the value 4, each with probability $\frac{1}{2}$. Consider a **call option** with strike $K = 6$: this is the right, but not the obligation, to buy one share at time 1 for the price 6. If $S_1 = 7$ we exercise the right and make 1; if $S_1 = 4$ we do not, and make 0. So the payout is

$$Y = (S_1 - K)^+ = \begin{cases} 1 & \text{if } S_1 = 7, \\ 0 & \text{if } S_1 = 4. \end{cases}$$

What is this option worth at time 0? A first guess might be $\mathbb{E}Y = \frac{1}{2}$. That guess is wrong, and the reason it is wrong is the whole point of the course.

Observe that $Y = \frac{1}{3}(S_1 - 4)$: indeed $\frac{1}{3}(7-4) = 1$ and $\frac{1}{3}(4-4) = 0$. So if at time 0 we buy $\frac{1}{3}$ of a share and borrow $\frac{4}{3}$ from the bank, our time-1 wealth is exactly $\frac{1}{3}S_1 - \frac{4}{3} = Y$, whatever happens. The cost of setting this up is

$$\tfrac{1}{3} \cdot 5 - \tfrac{4}{3} = \tfrac{5}{3} - \tfrac{4}{3} = \tfrac{1}{3}.$$

So a portfolio costing $\frac{1}{3}$ reproduces the option payout exactly. If the option were sold for more than $\frac{1}{3}$, we could sell the option, set up the portfolio, and pocket the difference with no risk whatsoever; if it were sold for less, we would do the reverse. So the only price consistent with the absence of a free lunch is $\frac{1}{3}$.

There is a second route to the same number. Suppose we pretend the probability of the up-move is q rather than $\frac{1}{2}$, chosen so that the share price is 'fair', i.e. $\mathbb{E}_{\mathbb{Q}}[S_1] = (1+r)S_0 = 5$. Then $7q + 4(1-q) = 5$, giving $q = \frac{1}{3}$. And under this measure,

$$\frac{\mathbb{E}_{\mathbb{Q}}[Y]}{1+r} = \tfrac{1}{3} \cdot 1 + \tfrac{2}{3} \cdot 0 = \tfrac{1}{3}.$$

The same answer. That this is not a coincidence is the content of the fundamental theorem of asset pricing, which we prove in §4.

Notice that the true probability $p = \frac{1}{2}$ played no role at all in the answer. This is the single most counterintuitive feature of derivative pricing, and it is worth being clear about why it happens: the option payout is a *deterministic function* of the share price, and the share price is already trading at a known price. All the information about probabilities that we need is already baked into $S_0 = 5$.

§2 Mean-Variance Analysis

Our agent has to choose a portfolio θ. On what basis? The oldest quantitative answer, due to Markowitz in 1952, is: get the expected return you want, and take as little risk as possible along the way, where 'risk' means variance.

§2.1 The Markowitz Problem

Definition 2.1 (Mean-variance portfolio problem)
Given initial wealth $X_0 = x$ and a target mean $m \in \mathbb{R}$, the **mean-variance portfolio problem** is to

$$\text{minimise } \operatorname{Var}(X_1) \quad \text{subject to} \quad \mathbb{E}[X_1] = m,$$

over portfolios $\theta \in \mathbb{R}^d$.

Throughout this section we assume that S_1 is square integrable, and we write

$$\mu = \mathbb{E}[S_1], \qquad V = \operatorname{Cov}(S_1) = \mathbb{E}\big[(S_1 - \mu)(S_1 - \mu)^\top\big].$$

The matrix V is always non-negative definite; we assume in addition that it is *positive* definite, so that V^{-1} exists. (If it is not, some portfolio of the risky assets is riskless, and one should first remove the redundancy.) We also assume $\mu \neq (1+r)S_0$, since otherwise the risky assets offer no expected reward over the bank and the problem is uninteresting.

Since $X_1 = (1+r)x + \theta^\top \xi$ with $\xi = S_1 - (1+r)S_0$, we have

$$\mathbb{E}[X_1] = (1+r)x + \theta^\top(\mu - (1+r)S_0), \qquad \mathrm{Var}(X_1) = \theta^\top V \theta.$$

So the problem is: minimise the quadratic form $\theta^\top V\theta$ subject to the affine constraint $\theta^\top(\mu - (1+r)S_0) = m - (1+r)x$. This is a problem you could do with Lagrange multipliers, but there is a slicker argument which also delivers uniqueness for free.

Theorem 2.2 (Markowitz)

Write $\nu = \mu - (1+r)S_0$. The mean-variance portfolio problem has the unique solution

$$\theta^* = \lambda V^{-1}\nu, \qquad \lambda = \frac{m-(1+r)x}{\nu^\top V^{-1}\nu}.$$

Proof. First, θ^* is feasible: $(\theta^*)^\top \nu = \lambda \nu^\top V^{-1}\nu = m - (1+r)x$, as required.

Now let θ be any other feasible portfolio and set $\epsilon = \theta - \theta^* \neq 0$. Expanding,

$$\theta^\top V\theta = (\theta^*)^\top V\theta^* + 2\epsilon^\top V\theta^* + \epsilon^\top V\epsilon.$$

The cross term vanishes: since both θ and θ^* are feasible, $\epsilon^\top\nu = 0$, and hence

$$\epsilon^\top V\theta^* = \lambda\,\epsilon^\top V V^{-1}\nu = \lambda\,\epsilon^\top\nu = 0.$$

As V is positive definite and $\epsilon \neq 0$ we get $\epsilon^\top V\epsilon > 0$, so $\theta^\top V\theta > (\theta^*)^\top V\theta^*$. Hence θ^* is the unique minimiser. □

The vector $V^{-1}\nu$ appearing here is important enough to be named.

Definition 2.3 (Market portfolio)

The **market portfolio** is $\theta^{\mathrm{mar}} = V^{-1}\big(\mu-(1+r)S_0\big)$, and we write $S_t^{\mathrm{mar}} = (\theta^{\mathrm{mar}})^\top S_t$ for its price.

Corollary 2.4 (Mutual fund theorem)

A portfolio is mean-variance efficient if and only if it is a scalar multiple of the market portfolio.

Proof. Immediate from Theorem 2.2: the optimal θ^* is $\lambda\theta^{\mathrm{mar}}$, and as m ranges over $\mathbb{R}$, λ ranges over $\mathbb{R}$. □

This is a genuinely striking conclusion. Every investor, no matter how risk-hungry or risk-shy, should hold the risky assets *in the same proportions*; the only thing that distinguishes them is how much they put in the bank. In practice this is the theoretical justification for index funds: if the theory holds, you need exactly two products, a bank account and one fund.

§2.2 The Efficient Frontier

As θ ranges over $\mathbb{R}^d$, the pair $(\mathrm{Var}(X_1), \mathbb{E}[X_1])$ traces out a subset of the right half plane. Its left-hand boundary is the object of interest.

Definition 2.5 (Efficient frontier)

The **mean-variance efficient frontier** is the set of points

$$\left\{ \big(\min\{\mathrm{Var}(X_1) : \mathbb{E}[X_1] = m\},\, m \big) : m \in \mathbb{R} \right\}.$$

A portfolio achieving a point of the frontier is called **mean-variance efficient**.

Corollary 2.6 (Shape of the frontier)

In the (v, m) plane the efficient frontier is the parabola

$$v = \frac{\big(m - (1+r)x\big)^2}{\nu^\top V^{-1} \nu}, \qquad \nu = \mu - (1+r)S_0.$$

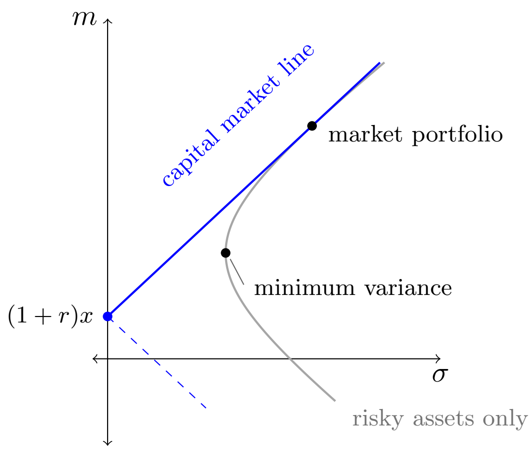
Proof. Substituting $\theta^* = \lambda V^{-1}\nu$ into $v = \theta^\top V \theta$ gives $v = \lambda^2 \nu^\top V^{-1} \nu$, and $\lambda = (m - (1+r)x)/(\nu^\top V^{-1}\nu)$. □

It is more usual, and more informative, to plot the frontier in the (σ, m) plane, where $\sigma = \mathrm{sd}(X_1) = \sqrt{v}$ is the standard deviation. Taking square roots in Corollary 2.6, the frontier becomes a pair of *straight lines* through the point $(0, (1+r)x)$, of slopes $\pm\sqrt{\nu^\top V^{-1}\nu}$. Only the upper line is of any interest – given a level of risk one wants the larger mean – and it is called the **capital market line**. Its slope

$$\frac{m - (1+r)x}{\sigma} = \sqrt{\nu^\top V^{-1}\nu}$$

is the **Sharpe ratio**, the amount of excess expected return purchased per unit of risk.

If we forbid the agent from using the bank account, and force them to hold only risky assets, then the corresponding frontier is instead a hyperbola in the (σ, m) plane, with a leftmost point (the **minimum-variance portfolio**). The capital market line is tangent to that hyperbola, and the point of tangency is exactly the market portfolio, normalised to cost x. The picture to hold in your head is the following.



The dashed part of the capital market line corresponds to $\lambda < 0$, i.e. to shorting the market portfolio; those portfolios solve the minimisation problem but no sensible agent would choose them.

Corollary 2.7

The problem of minimising $\mathrm{Var}(X_1)$ subject to $\mathbb{E}[X_1] \geq m$ has the same solution as the mean-variance problem whenever $m \geq (1+r)x$. If $m < (1+r)x$, the solution is $\theta = 0$.

Proof. If $m \geq (1+r)x$ then the constraint binds, since variance is increasing in $|\lambda|$ and the mean is increasing in λ. If $m < (1+r)x$ then $\theta = 0$ gives $\mathrm{Var}(X_1) = 0$ and $\mathbb{E}[X_1] = (1+r)x \geq m$, which is clearly optimal. □

Remark. Variance is a strange measure of risk. It punishes unexpectedly *good* outcomes exactly as much as unexpectedly bad ones, and it takes no account of the shape of the tail. We will replace it with something better in §3. On the other hand, mean-variance analysis has the enormous advantage of being solvable in closed form, which is why it is still used.

§2.3 The Capital Asset Pricing Model

We now ask an equilibrium question: if *everyone* behaves as in the previous section, what do prices look like? The answer is the capital asset pricing model of Sharpe (1964), and its content is that the expected excess return of any asset is determined by a single number, its exposure to the market as a whole.

We begin with an entirely elementary fact, which is just linear regression.

Proposition 2.8

Let X, Y be square integrable random variables with $\mathrm{Var}(X) > 0$. Then there are unique constants α, β such that

$$Y = \alpha + \beta X + Z, \qquad \mathbb{E}[Z] = 0, \quad \mathrm{Cov}(Z, X) = 0.$$

Explicitly, $\beta = \mathrm{Cov}(X,Y)/\mathrm{Var}(X)$ and $\alpha = \mathbb{E}[Y] - \beta\,\mathbb{E}[X]$.

Proof. Setting $Z = Y - \alpha - \beta X$, the two conditions read

$$0 = \mathbb{E}[Y] - \alpha - \beta\mathbb{E}[X], \qquad 0 = \mathrm{Cov}(Y, X) - \beta\,\mathrm{Var}(X),$$

which have the unique solution stated. □

Now let us specialise to the market portfolio. Recall $S_t^{\mathrm{mar}} = (\theta^{\mathrm{mar}})^\top S_t$ and write $\xi = S_1 - (1+r)S_0$, $\xi^{\mathrm{mar}} = S_1^{\mathrm{mar}} - (1+r)S_0^{\mathrm{mar}} = (\theta^{\mathrm{mar}})^\top \xi$.

Theorem 2.9

Let $B = \mathrm{Cov}(S_1, S_1^{\mathrm{mar}})/\mathrm{Var}(S_1^{\mathrm{mar}}) \in \mathbb{R}^d$ and set $Z = \xi - B\,\xi^{\mathrm{mar}}$. Then $\mathbb{E}[Z] = 0$ and $\mathrm{Cov}(Z, S_1^{\mathrm{mar}}) = 0$.

Proof. Write $\nu = \mu - (1+r)S_0 = \mathbb{E}[\xi]$, so $\theta^{\mathrm{mar}} = V^{-1}\nu$. Then

$$\mathbb{E}[\xi^{\mathrm{mar}}] = (\theta^{\mathrm{mar}})^\top \nu = \nu^\top V^{-1} \nu = (\theta^{\mathrm{mar}})^\top V \theta^{\mathrm{mar}} = \mathrm{Var}(S_1^{\mathrm{mar}}),$$

using $V\theta^{\text{mar}} = \nu$. On the other hand

$$\operatorname{Cov}(S_1, S_1^{\text{mar}}) = \operatorname{Cov}(S_1, S_1)\theta^{\text{mar}} = VV^{-1}\nu = \nu = \mathbb{E}[\xi].$$

Therefore $B = \mathbb{E}[\xi]/\mathbb{E}[\xi^{\text{mar}}]$, whence $\mathbb{E}[Z] = \mathbb{E}[\xi] - B\,\mathbb{E}[\xi^{\text{mar}}] = 0$. Similarly,

$$\operatorname{Cov}(Z, S_1^{\text{mar}}) = \operatorname{Cov}(S_1, S_1^{\text{mar}}) - B\operatorname{Var}(S_1^{\text{mar}}) = \nu - \nu = 0. \qquad \square$$

To turn this into an economic statement we need the market to clear. Suppose asset i exists in total supply n^i, write $n = (n^1, \dots, n^d)^\top$, and define the **index return**

$$R^{\text{index}} = \frac{n^\top S_1}{n^\top S_0} - 1,$$

the return on holding the entire market in proportion to supply. Let $R^i = S_1^i/S_0^i - 1$ be the return on asset i.

Axiom 2.10 (CAPM assumptions)

There are K investors, with portfolios $\theta^1, \dots, \theta^K$ satisfying **market clearing** $\sum_k \theta^k = n$. Every investor agrees on the values of $\mu = \mathbb{E}[S_1]$ and $V = \operatorname{Cov}(S_1)$ (*uniform beliefs*) and every investor holds a mean-variance efficient portfolio, though possibly with different target means (*uniform preferences*).

Theorem 2.11 (Capital asset pricing model)

Under the above assumptions, write the regression

$$R^i - r = \alpha^i + \beta^i\big(R^{\text{index}} - r\big) + \epsilon^i, \qquad \mathbb{E}[\epsilon^i] = 0,\ \operatorname{Cov}(R^{\text{index}}, \epsilon^i) = 0,$$

as in Proposition 2.8. Then $\alpha^i = 0$ for every i.

Proof. By the mutual fund theorem each investor's portfolio is $\theta^k = \lambda_k\theta^{\text{mar}}$ for some $\lambda_k \in \mathbb{R}$. Market clearing gives $n = \big(\sum_k \lambda_k\big)\theta^{\text{mar}}$, so n is a positive multiple of the market portfolio, and therefore

$$R^{\text{index}} = \frac{n^\top S_1}{n^\top S_0} - 1 = \frac{S_1^{\text{mar}}}{S_0^{\text{mar}}} - 1.$$

Now apply Theorem 2.9 coordinatewise. Dividing $\xi^i = B^i\xi^{\text{mar}} + Z^i$ by S_0^i,

$$R^i - r = \frac{1}{S_0^i}\big(S_1^i - (1+r)S_0^i\big) = \frac{B^i S_0^{\text{mar}}}{S_0^i}\big(R^{\text{index}} - r\big) + \frac{Z^i}{S_0^i}.$$

Since $\mathbb{E}[Z^i] = 0$ and $\operatorname{Cov}(Z^i, R^{\text{index}}) = 0$, uniqueness in Proposition 2.8 identifies $\beta^i = B^i S_0^{\text{mar}}/S_0^i$, $\epsilon^i = Z^i/S_0^i$ and $\alpha^i = 0$. $\square$

So under CAPM the expected excess return on asset i is

$$\mathbb{E}[R^i] - r = \beta^i\big(\mathbb{E}[R^{\text{index}}] - r\big),$$

and nothing else about asset i matters. The quantity β^i is a genuinely used piece of market vocabulary. An asset with $\beta = 1.5$ tends to move one and a half times as much as the market; an asset with $\beta = 0$ is uncorrelated with the market and, however volatile it may be on its own, commands no excess return at all, because its risk can be diversified away.

Remark. It is worth being honest about the assumptions. Uniform beliefs is not plausible: if everyone agreed on μ and V there would be very little trading. Uniform preferences is worse: we have already noted that variance is a poor proxy for risk. And empirically, when one regresses returns against the index, the estimated α's are not zero – searching for assets with positive α is more or less the definition of active management. Nonetheless CAPM is the standard baseline against which such claims are measured.

§3 Utility and Risk Aversion

Mean-variance analysis fixes the mean and minimises variance. This is unsatisfactory in two ways: it does not let an agent trade off return against risk, and variance is the wrong notion of risk. We now set up a framework which does both.

§3.1 The Expected Utility Hypothesis

Definition 3.1 (Expected utility hypothesis)

Every agent has a **utility function** $U : \mathbb{R} \to \mathbb{R}$ (or defined on an interval), and prefers a random payout X to a random payout Y – written $X \succ Y$ – if and only if

$$\mathbb{E}[U(X)] > \mathbb{E}[U(Y)].$$

If $\mathbb{E}[U(X)] = \mathbb{E}[U(Y)]$ the agent is **indifferent** between them, written $X \sim Y$.

Two immediate observations. First, if $\tilde{U} = aU + b$ with $a > 0$, then $\tilde{U}$ induces exactly the same preferences as U; utility functions are only defined up to positive affine transformations, and so ‘utility’ is not a quantity one can meaningfully compare across agents. Second, the criterion depends on X only through its *distribution*, so we may equally think of preferences as an ordering on distribution functions.

Historically the hypothesis was introduced by Daniel Bernoulli to resolve the following puzzle.

Example 3.2 (The St. Petersburg paradox)

A fair coin is tossed until it first comes up heads. If this takes n tosses, the casino pays you 2^n pounds. What should you pay to play?

The expected payout is

$$\sum_{n\geq 1} 2^{-n} \cdot 2^n = \sum_{n\geq 1} 1 = \infty,$$

so if one maximises expected wealth one should be willing to pay any finite entry fee. Almost nobody would pay even £100. Bernoulli’s resolution: an agent maximises not expected wealth but expected utility, and if $U(x) = \log x$ then the expected utility of the game is

$$\sum_{n\geq 1} 2^{-n}\log(2^n) = \log 2 \sum_{n\geq 1} n2^{-n} = 2\log 2 = \log 4,$$

so the game is worth exactly £4 to a logarithmic agent. The moral is that a fixed amount of money is worth less to you when you are rich.

A more principled motivation is a representation theorem. Suppose an agent has some preference relation $\succ$ on distribution functions, about which we assume only the following.

Definition 3.3 (von Neumann–Morgenstern axioms)

A preference relation $\succ$ on distribution functions satisfies the **von Neumann–Morgenstern axioms** if

(i) (*Completeness*) for all F, G, exactly one of $F \succ G$, $G \succ F$, $F \sim G$ holds;

(ii) (*Transitivity*) if $F \succ G$ and $G \succ H$ then $F \succ H$;

(iii) (*Independence*) if $F \succ G$ and $p \in [0,1]$ then $pF + (1-p)H \succ pG + (1-p)H$ for every H;

(iv) (*Continuity*) if $F \succ G \succ H$ then there is $p \in [0,1]$ with $G \sim pF + (1-p)H$.

Theorem 3.4 (von Neumann–Morgenstern)

If $\succ$ satisfies the above axioms then there exists a function U such that

$$F \succ G \iff \int U \, \mathrm{d}F > \int U \, \mathrm{d}G.$$

We shall not prove this. The point is that the expected utility hypothesis is not an arbitrary modelling choice: it is forced on us by a short list of conditions each of which looks, at first glance, like a requirement of bare rationality. (The independence axiom is the controversial one, and there is a well-developed literature of experiments in which people violate it.)

§3.2 Concavity, Jensen and the Certainty Equivalent

We will almost always assume that U is increasing, and very often that it is concave.

Definition 3.5

A utility function U is **increasing** if $X > Y$ almost surely implies $\mathbb{E}[U(X)] > \mathbb{E}[U(Y)]$, and is **concave** if

$$U(px + (1-p)y) \geq pU(x) + (1-p)U(y) \qquad \text{for all } x, y \in \mathbb{R},\ p \in [0,1].$$

An agent with a concave utility function is called **risk averse**.

If U is twice differentiable then increasing means $U' > 0$ and concave means $U'' \leq 0$. In words: more money is better, but each additional pound is worth slightly less than the one before.

The connection between concavity and dislike of risk is Jensen's inequality, which you met in Part IA Probability.

Proposition 3.6

If U is concave and X is integrable then $\mathbb{E}[U(X)] \leq U(\mathbb{E}[X])$. Hence a risk-averse agent weakly prefers the certain payout $\mathbb{E}[X]$ to the random payout X.

This suggests a way of measuring *how much* an agent dislikes a given gamble.

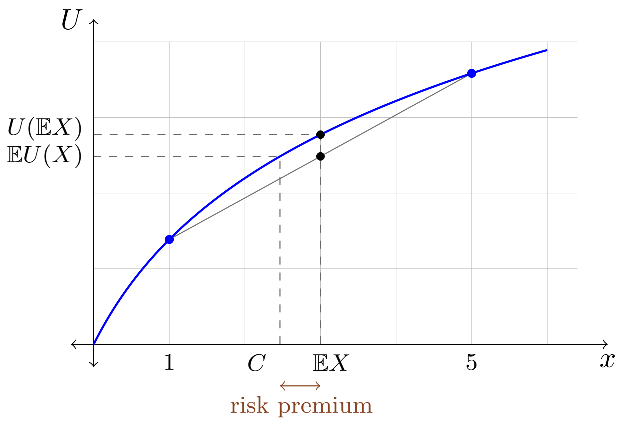
Definition 3.7 (Certainty equivalent)

Let U be increasing and continuous and let X be a payout. The **certainty equivalent** of X is the unique $C(X) \in \mathbb{R}$ with

$$U(C(X)) = \mathbb{E}[U(X)],$$

i.e. the sure amount which the agent values exactly as highly as the gamble. The **risk premium** is $\mathbb{E}[X] - C(X)$.

By Jensen, a risk-averse agent has $C(X) \leq \mathbb{E}[X]$, so the risk premium is non-negative: it is the discount the agent demands for bearing the risk. The whole construction is visible in a single picture. Take $U(x) = \log(1 + x)$ and let X take the values 1 and 5 with probability $\frac{1}{2}$ each, so $\mathbb{E}[X] = 3$.



The chord joining the two outcomes lies below the graph; its midpoint has height $\mathbb{E}[U(X)]$, and reading horizontally across to the curve and then down gives the certainty equivalent $C \approx 2.46$. The gap between C and $\mathbb{E}X = 3$ is the risk premium, and it is exactly the vertical gap between the chord and the curve, transported through U. A more concave utility bends the curve further away from the chord and so demands a larger premium.

§3.3 Arrow–Pratt Risk Aversion

To make 'more concave' precise we want a number. The obvious candidate $-U''$ is no good, because U is only defined up to positive affine transformations and $-U''$ is not invariant under $U \mapsto aU + b$. Dividing by U' fixes this.

Definition 3.8 (Arrow–Pratt coefficients)

Let U be twice differentiable with $U' > 0$. The **coefficient of absolute risk aversion** at x is

$$A(x) = -\frac{U''(x)}{U'(x)},$$

and the **coefficient of relative risk aversion** at x is

$$R(x) = -\frac{xU''(x)}{U'(x)} = xA(x).$$

Both are invariant under $U \mapsto aU + b$ for $a > 0$, as they should be. The interpretation is the following: if an agent with wealth x is offered a small gamble with mean zero and variance σ^2, then a second-order Taylor expansion gives

$$\mathbb{E}[U(x+\epsilon)] \approx U(x) + \tfrac{1}{2}U''(x)\sigma^2, \qquad U(x-\pi) \approx U(x) - U'(x)\pi,$$

so the risk premium is $\pi \approx \frac{1}{2}A(x)\sigma^2$. Absolute risk aversion measures the premium demanded for a gamble of a fixed size; relative risk aversion measures the premium for a gamble proportional to wealth.

Two families of utility functions are singled out by asking for these coefficients to be constant.

Example 3.9 (CARA utility)

A utility with **constant absolute risk aversion** $\gamma > 0$ satisfies $-U''/U' = \gamma$, so $(\log U')' = -\gamma$ and $U'(x) = Ce^{-\gamma x}$. Integrating,

$$U(x) = -\frac{C}{\gamma}e^{-\gamma x},$$

and normalising, 'the' CARA utility is $U(x) = -e^{-\gamma x}$, defined on all of $\mathbb{R}$.

Example 3.10 (CRRA utility)

A utility with **constant relative risk aversion** $R > 0$ satisfies $-xU''/U' = R$, so $(\log U')' = -R/x$ and $U'(x) = Cx^{-R}$ for $x > 0$. If $R \neq 1$, integrating gives

$$U(x) = \frac{x^{1-R}}{1-R}, \qquad x > 0,$$

up to affine equivalence. If $R = 1$ the integral produces a logarithm, and we obtain the **logarithmic utility** $U(x) = \log x$, $x > 0$.

Remark. CARA utility has the convenient property that the optimal *amount of money* invested in the risky assets does not depend on wealth; CRRA utility has the property that the optimal *proportion* of wealth invested does not. The latter is closer to how people appear to behave, and CRRA utilities are also the ones that combine well with lognormal models, since $\mathbb{E}[e^{(1-R)Z}]$ is computable for Gaussian Z. Note that CRRA utility with $R > 1$ has $U(x) \to -\infty$ as $x \downarrow 0$, so such an agent will never accept a strategy which can bankrupt them.

§3.4 The Utility Maximisation Problem

We can now pose the central optimisation problem of the one-period theory: given a utility U and initial wealth x,

$$\text{maximise } \mathbb{E}[U(X_1)] \quad \text{over } X_1 \in \mathcal{X}(x),$$

that is, maximise

$$f(\theta) := \mathbb{E}\Big[U\big((1+r)x + \theta^\top \xi\big)\Big] \quad \text{over } \theta \in \mathbb{R}^d.$$

Theorem 3.11 (First-order condition)
Suppose U is differentiable and θ^* is optimal, with $X_1^* = (1+r)x + (\theta^*)^\top \xi$. Suppose further that differentiation under the expectation is justified. Then

$$\mathbb{E}\big[U'(X_1^*)\,\xi\big] = 0, \qquad \text{i.e.} \qquad \mathbb{E}\big[U'(X_1^*)(S_1 - (1+r)S_0)\big] = 0.$$

Proof. The function f is differentiable with $\nabla f(\theta) = \mathbb{E}[U'((1+r)x+\theta^\top \xi)\,\xi]$, provided we may exchange ∇ and $\mathbb{E}$; this is justified by dominated convergence when, for instance, U is concave and differentiable and $U((1+r)x + \theta^\top \xi)$ is integrable for all θ in a neighbourhood of θ^*. Since θ^* is an interior maximum (there are no constraints), $\nabla f(\theta^*) = 0$. □

Note that if U is concave then f is concave, so the first-order condition is sufficient as well as necessary. The equation says that the marginal utility of the optimal wealth is 'orthogonal' to every excess return, which is exactly the statement that no small perturbation of the portfolio helps.

§3.5 State Price Densities

The first-order condition can be rewritten in a way that removes all mention of the agent, and this turns out to be the key to everything that follows.

Definition 3.12 (State price density)
A **state price density** (or **pricing kernel**) is an almost surely positive random variable Z with

$$\mathbb{E}[Z] = \frac{1}{1+r} \qquad \text{and} \qquad \mathbb{E}[ZS_1] = S_0.$$

The name 'pricing kernel' is transparent: multiplying by Z and taking expectations converts time-1 prices into time-0 prices. Note the first condition is just the second applied to the bank account, whose time-1 value is the constant $1 + r$.

To see why it is called a *state price* density, consider a market on $\Omega = \{\omega_1, \ldots, \omega_d\}$ in which asset i is the **Arrow–Debreu security** paying $S_1^i = \mathbb{1}_{\{\omega_i\}}$, i.e. one pound if state i occurs and nothing otherwise. Then $\mathbb{E}[ZS_1^i] = Z(\omega_i)\mathbb{P}(\{\omega_i\}) = S_0^i$, so

$$Z(\omega_i) = \frac{S_0^i}{\mathbb{P}(\{\omega_i\})}.$$

So Z is the price of a pound delivered in state ω_i, per unit of probability – a density of state prices with respect to $\mathbb{P}$.

Theorem 3.13

In the setting of Theorem 3.11, suppose additionally that $U' > 0$ and $\mathbb{E}[U'(X_1^*)] < \infty$. Then

$$Z = \frac{U'(X_1^*)}{(1+r)\,\mathbb{E}[U'(X_1^*)]}$$

is a state price density.

Proof. Certainly $Z > 0$ and $\mathbb{E}[Z] = 1/(1+r)$. For the second condition, Theorem 3.11 gives $\mathbb{E}[U'(X_1^*)(S_1 - (1+r)S_0)] = 0$, i.e.

$$\mathbb{E}[U'(X_1^*)S_1] = (1+r)S_0\,\mathbb{E}[U'(X_1^*)].$$

Dividing by $(1+r)\mathbb{E}[U'(X_1^*)]$ gives $\mathbb{E}[ZS_1] = S_0$. □

So *if* an agent with an increasing utility can solve their optimisation problem, *then* a state price density exists. This is our first sight of the fundamental theorem: the existence of a solution to some agent's problem forces a consistency condition on the market. In §4 we will strip away the agent entirely and find that the correct hypothesis is simply the absence of arbitrage.

§4 Arbitrage and the Fundamental Theorem

We now come to the heart of the course. The idea is to eliminate the agent altogether: rather than asking what a particular investor with a particular utility would pay, we ask what prices are consistent with the impossibility of a free lunch. Remarkably, this is enough to determine prices exactly in a wide class of models.

§4.1 Equivalent Measures

We first need a small amount of measure-theoretic machinery for changing probability measures.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let Y be an almost surely positive random variable with $\mathbb{E}_{\mathbb{P}}[Y] = 1$. Define

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}\big[Y \mathbb{1}_A\big], \qquad A \in \mathcal{F}.$$

Then $\mathbb{Q}(\Omega) = 1$ and countable additivity follows from monotone convergence, so $\mathbb{Q}$ is a probability measure. Moreover, since $Y > 0$ almost surely, $\mathbb{Q}(A) = 0$ if and only if $\mathbb{P}(A) = 0$.

Definition 4.1 (Equivalent measures)

Two probability measures $\mathbb{P}, \mathbb{Q}$ on $(\Omega, \mathcal{F})$ are **equivalent**, written $\mathbb{P} \sim \mathbb{Q}$, if they have the same null sets: $\mathbb{P}(A) = 0 \iff \mathbb{Q}(A) = 0$.

Equivalent measures agree on what is possible and what is impossible; they disagree about how likely the possible things are. Every construction above produces an equivalent measure, and the Radon–Nikodym theorem says that these are the only ones.

Theorem 4.2 (Radon–Nikodym)
$\mathbb{P}$ and $\mathbb{Q}$ are equivalent if and only if there is an almost surely positive random variable Y, unique up to null sets, with $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}[Y \mathbb{1}_A]$ for all $A \in \mathcal{F}$.

We write $Y = \mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ and call it the **Radon–Nikodym derivative**, or the **likelihood ratio** – a name which should be familiar from Part IB Statistics, where exactly the same object appears in the Neyman–Pearson lemma. The essential computational fact is that for $\mathbb{Q}$-integrable Z,

$$\mathbb{E}_{\mathbb{Q}}[Z] = \mathbb{E}_{\mathbb{P}}\left[\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} Z\right],$$

which holds for indicators by definition and in general by the usual approximation argument.

Example 4.3
If $\Omega = \{\omega_1, \ldots, \omega_n\}$ is finite and $\mathbb{P}(\{\omega_i\}) > 0$ for each i, then $\mathbb{P} \sim \mathbb{Q}$ exactly when $\mathbb{Q}(\{\omega_i\}) > 0$ for each i, and

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(\omega_i) = \frac{\mathbb{Q}(\{\omega_i\})}{\mathbb{P}(\{\omega_i\})}.$$

Example 4.4 (Changing the mean of a Gaussian)
Let $X \sim N(\mu, \sigma^2)$ under $\mathbb{P}$ and set $Y = e^{X-\mu-\sigma^2/2}$. Then $Y > 0$ and, using the moment generating function of the normal, $\mathbb{E}_{\mathbb{P}}[Y] = e^{-\mu-\sigma^2/2}\mathbb{E}_{\mathbb{P}}[e^X] = 1$. Let $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = Y$. For bounded measurable f,

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}[f(X)] &= \mathbb{E}_{\mathbb{P}}\left[e^{X-\mu-\sigma^2/2} f(X)\right] \\
&= \int_{\mathbb{R}} e^{x-\mu-\sigma^2/2} f(x) \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}\, \mathrm{d}x \\
&= \int_{\mathbb{R}} f(x) \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu-\sigma^2)^2/2\sigma^2}\, \mathrm{d}x,
\end{aligned}$$

on completing the square. So under $\mathbb{Q}$ we have $X \sim N(\mu + \sigma^2, \sigma^2)$: the change of measure has shifted the mean and left the variance alone. This little computation is the seed of the Cameron–Martin theorem in §7, and hence of the entire Black–Scholes derivation.

§4.2 Risk-Neutral Measures

Definition 4.5 (Risk-neutral measure)
A **risk-neutral measure** (or **equivalent martingale measure**) for the one-period market is a probability measure $\mathbb{Q} \sim \mathbb{P}$ with

$$\mathbb{E}_{\mathbb{Q}}\left[\frac{S_1}{1+r}\right] = S_0.$$

Under $\mathbb{Q}$, every asset has the same expected return as the bank account, so a $\mathbb{Q}$-agent is indifferent between them – hence 'risk neutral'. The connection with state price densities is immediate.

Proposition 4.6

$\mathbb{Q}$ is a risk-neutral measure if and only if $Z = \frac{1}{1+r}\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$ is a state price density. In particular, state price densities and risk-neutral measures are in bijection.

Proof. If $\mathbb{Q} \sim \mathbb{P}$ with density Y, then $Z = Y/(1+r) > 0$ and $\mathbb{E}_{\mathbb{P}}[Z] = 1/(1+r)$ automatically. Furthermore $\mathbb{E}_{\mathbb{P}}[ZS_1] = \frac{1}{1+r}\mathbb{E}_{\mathbb{Q}}[S_1]$, which equals S_0 precisely when $\mathbb{Q}$ is risk-neutral. Conversely, given a state price density Z, put $Y = (1+r)Z$; then $Y > 0$ and $\mathbb{E}_{\mathbb{P}}[Y] = 1$, so Y defines an equivalent measure $\mathbb{Q}$, which is risk-neutral by the same computation. $\square$

Example 4.7 (The one-period binomial model)

Take $d = 1$ and suppose

$$\mathbb{P}\big(S_1 = (1+b)S_0\big) = p = 1 - \mathbb{P}\big(S_1 = (1+a)S_0\big), \qquad a < b,\ p \in (0,1).$$

A measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ is determined by $q = \mathbb{Q}(S_1 = (1+b)S_0) \in (0,1)$, and risk-neutrality demands

$$q(1+b)S_0 + (1-q)(1+a)S_0 = (1+r)S_0 \implies q = \frac{r-a}{b-a}.$$

This lies in $(0,1)$ if and only if $a < r < b$. So a risk-neutral measure exists exactly when the bank rate lies strictly between the two possible returns on the stock, and then it is unique. Both halves of this statement will be explained by the theorems below.

§4.3 Arbitrage

Definition 4.8 (Arbitrage)

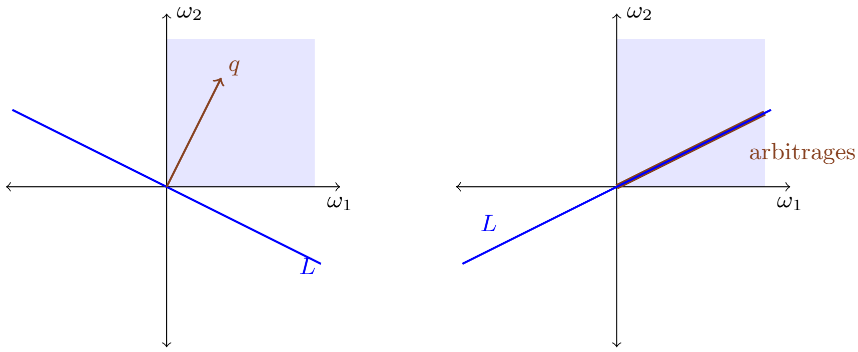
A portfolio $\phi \in \mathbb{R}^d$ is an **arbitrage** if

$$\phi^\top \xi \geq 0 \ \text{ a.s.} \qquad \text{and} \qquad \mathbb{P}(\phi^\top \xi > 0) > 0,$$

where $\xi = S_1 - (1+r)S_0$. A market with no arbitrage is called **arbitrage-free**.

An arbitrage is a portfolio which costs nothing over and above what the bank would have given you, never loses, and sometimes wins: a free lunch. Note that the definition involves neither a utility function nor an initial wealth, and depends on $\mathbb{P}$ only through its null sets. In particular equivalent measures have exactly the same arbitrages.

Here is the geometry, in the simplest possible case: $d = 1$ and two states ω_1, ω_2. Identify a random variable with the point of $\mathbb{R}^2$ recording its values in the two states. The attainable excess payoffs form the line $L = \{\theta\xi : \theta \in \mathbb{R}\}$ through the origin, and an arbitrage is a point of L lying in the closed positive quadrant but not at the origin.



no arbitrage arbitrage

On the left the line meets the shaded quadrant only at the origin, and there is a vector q with strictly positive entries orthogonal to L; normalising q to be a probability vector produces exactly a risk-neutral measure. On the right the line runs into the interior of the quadrant, and every point of the thick segment is an arbitrage; no strictly positive vector can be orthogonal to L. The fundamental theorem is the assertion that this dichotomy is completely general.

Before proving it, let us record the link with the previous section.

Proposition 4.9

If the market admits an arbitrage, then no increasing utility function admits an optimal portfolio.

Proof. Let ϕ be an arbitrage and θ any portfolio, with $X_1 = (1+r)x + \theta^\top \xi$ and $\tilde{X}_1 = (1+r)x + (\theta+\phi)^\top \xi$. Then $\tilde{X}_1 \geq X_1$ almost surely, with strict inequality on an event of positive probability. Since U is increasing, $\mathbb{E}[U(\tilde{X}_1)] > \mathbb{E}[U(X_1)]$, so θ was not optimal. □

Combining Proposition 4.9 with Theorem 3.13: if some agent can solve their optimisation problem then there is no arbitrage, and there is a state price density. The fundamental theorem removes the agent from this chain of reasoning.

§4.4 The Fundamental Theorem of Asset Pricing

Theorem 4.10 (Fundamental theorem of asset pricing)

The one-period market admits no arbitrage if and only if there exists a risk-neutral measure.

Proof. ($\Leftarrow$) Suppose $\mathbb{Q}$ is risk-neutral and ϕ is an arbitrage. Since $\mathbb{P} \sim \mathbb{Q}$ we have $\phi^\top \xi \geq 0$ $\mathbb{Q}$-almost surely too. But risk-neutrality says $\mathbb{E}_{\mathbb{Q}}[\xi] = \mathbb{E}_{\mathbb{Q}}[S_1] - (1+r)S_0 = 0$, so

$$\mathbb{E}_{\mathbb{Q}}[\phi^\top \xi] = \phi^\top \mathbb{E}_{\mathbb{Q}}[\xi] = 0.$$

A non-negative random variable with zero expectation vanishes almost surely, so $\phi^\top \xi = 0$ $\mathbb{Q}$-a.s., hence $\mathbb{P}$-a.s., contradicting $\mathbb{P}(\phi^\top \xi > 0) > 0$.

($\Rightarrow$) Suppose there is no arbitrage; we construct a risk-neutral measure. We may make two harmless reductions. First, we may assume the components $\xi^1, \dots, \xi^d$ are linearly independent as random variables: otherwise, discard assets until they are, since a redundant asset contributes nothing new to $\{\theta^\top \xi\}$. Second, we may assume that $\mathbb{E}[e^{-\theta^\top \xi}] < \infty$ for every $\theta \in \mathbb{R}^d$: if not, replace $\mathbb{P}$ by the equivalent measure $\tilde{\mathbb{P}}$ with $\mathrm{d}\tilde{\mathbb{P}}/\mathrm{d}\mathbb{P} \propto e^{-\|\xi\|^2}$, which has the same arbitrages, and note that $e^{-\theta^\top \xi - \|\xi\|^2}$ is bounded.

Now consider

$$f(\theta) = \mathbb{E}\big[e^{-\theta^\top \xi}\big], \qquad \theta \in \mathbb{R}^d,$$

which is finite, strictly convex and smooth. Choose (θ_n) with $f(\theta_n) \to \inf_\theta f(\theta)$.

Case 1: (θ_n) has a bounded subsequence. By Bolzano–Weierstrass we may assume $\theta_n \to \theta_0$. Continuity of moment generating functions gives $f(\theta_0) = \inf_\theta f$, so θ_0 is a minimiser and

$$\nabla f(\theta_0) = -\mathbb{E}\big[e^{-\theta_0^\top \xi} \xi\big] = 0.$$

Setting $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = e^{-\theta_0^\top \xi}/\mathbb{E}[e^{-\theta_0^\top \xi}]$ defines an equivalent measure (the density is strictly positive) with $\mathbb{E}_{\mathbb{Q}}[\xi] = 0$, i.e. a risk-neutral measure.

Case 2: $\|\theta_n\| \to \infty$. We derive a contradiction. Put $\phi_n = \theta_n/\|\theta_n\|$; by compactness of the unit sphere we may assume $\phi_n \to \phi_0$ with $\|\phi_0\| = 1$. We claim $\phi_0^\top \xi \geq 0$ almost surely.

Suppose not, so that $\mathbb{P}(\phi_0^\top \xi < -\epsilon,\ \|\xi\| < R) > 0$ for some $\epsilon, R > 0$. For n large enough that $\|\phi_n - \phi_0\| \leq \epsilon/(2R)$, on the event $\{\phi_0^\top \xi < -\epsilon,\ \|\xi\| < R\}$ we have

$$\phi_n^\top \xi = (\phi_n - \phi_0)^\top \xi + \phi_0^\top \xi \leq \|\phi_n - \phi_0\|\, \|\xi\| + \phi_0^\top \xi \leq \tfrac{\epsilon}{2} - \epsilon = -\tfrac{\epsilon}{2}.$$

Since $f(\theta_n) \leq f(0) = 1$ for large n,

$$1 \geq \mathbb{E}\big[e^{-\|\theta_n\| \phi_n^\top \xi}\big] \geq \mathbb{E}\big[e^{-\|\theta_n\| \phi_n^\top \xi} \mathbb{1}_{\{\phi_0^\top \xi < -\epsilon, \|\xi\| < R\}}\big] \geq e^{\epsilon \|\theta_n\|/2}\, \mathbb{P}\big(\phi_0^\top \xi < -\epsilon,\ \|\xi\| < R\big).$$

The right-hand side tends to infinity as $n \to \infty$, which is absurd. Hence $\phi_0^\top \xi \geq 0$ a.s.

By the absence of arbitrage, $\phi_0^\top \xi \geq 0$ a.s. forces $\phi_0^\top \xi = 0$ a.s. But the ξ^i are linearly independent, so $\phi_0 = 0$, contradicting $\|\phi_0\| = 1$. So Case 2 cannot occur, and we are done. □

Remark. The proof is really a separating hyperplane argument in disguise. The convex function $f(\theta) = \mathbb{E}[e^{-\theta^\top \xi}]$ is a smooth surrogate for the indicator of the 'no free lunch' condition, and Case 2 is exactly the statement that if f has no minimiser then the recession direction ϕ_0 is an arbitrage. In finite sample spaces one can give a purely finite-dimensional proof using the separating hyperplane theorem, and the picture above is that proof.

§4.5 Contingent Claims and Payoff Diagrams

We now enlarge the market by an extra asset.

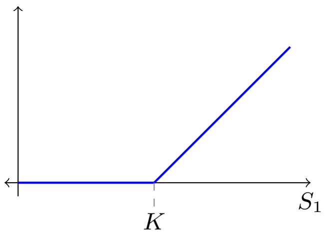
Definition 4.11 (Contingent claim)

A **contingent claim** with maturity 1 is a random variable Y, interpreted as a payout at time 1. It is **vanilla** if $Y = g(S_1)$ for some function g.

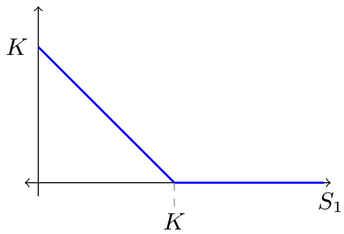
The two basic examples are the following.

Definition 4.12 (Call and put)
A **European call option** with **strike** K is the right, but not the obligation, to buy one unit of the asset at time 1 for the price K; its payout is $(S_1 - K)^+$. A **European put option** with strike K is the right to sell at price K, with payout $(K - S_1)^+$.

It is customary and extremely useful to draw the payout as a function of S_1.

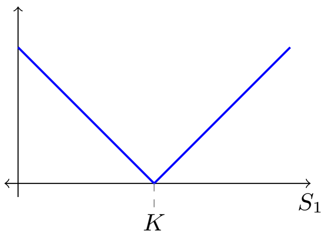


call, $(S_1 - K)^+$

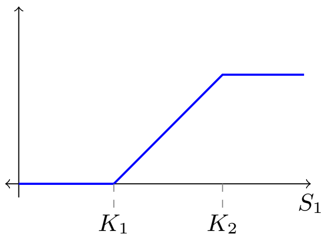


put, $(K - S_1)^+$

Because the space of claims is a vector space and pricing will turn out to be linear, one can build more elaborate payoffs by adding these together. Two standard combinations:



straddle, $|S_1 - K|$



bull spread

A **straddle** is a call plus a put with the same strike, and pays off when the price moves a long way in either direction: it is a bet on volatility rather than on direction. A **bull spread** is a long call at strike K_1 and a short call at strike $K_2 > K_1$; it is a bet that the price will rise, but with the upside capped in exchange for a lower initial cost. Notice that the bull spread payoff is bounded, which is often what a regulator wants to see.

§4.6 No-Arbitrage Pricing

Suppose the fundamental market (the d assets and the bank) is arbitrage-free, and we add a claim with payout Y and initial price π. When is the enlarged market still arbitrage-free?

Theorem 4.13 (No-arbitrage pricing)
Let the fundamental market be arbitrage-free. The augmented market containing the claim (Y, π) is arbitrage-free if and only if

$$\pi = \frac{1}{1+r}\mathbb{E}_{\mathbb{Q}}[Y]$$

for some risk-neutral measure $\mathbb{Q}$ of the fundamental market.

Proof. Apply Theorem 4.10 to the augmented market, whose price vectors are $(S_0, \pi)^\top$ and $(S_1, Y)^\top$. The augmented market is arbitrage-free if and only if there is $\mathbb{Q} \sim \mathbb{P}$ with

$$\frac{1}{1+r}\mathbb{E}_\mathbb{Q}\left[(S_1, Y)^\top\right] = (S_0, \pi)^\top.$$

The first d coordinates say exactly that $\mathbb{Q}$ is risk-neutral for the fundamental market, and the last says $\pi = \mathbb{E}_\mathbb{Q}[Y]/(1+r)$. □

Proposition 4.14

The set of no-arbitrage prices of a claim Y is an interval.

Proof. If $\mathbb{Q}_0, \mathbb{Q}_1$ are risk-neutral, so is $\mathbb{Q}_t = t\mathbb{Q}_1 + (1-t)\mathbb{Q}_0$ for $t \in [0,1]$: it is a probability measure, it is equivalent to $\mathbb{P}$ (its null sets are the common null sets), and $\mathbb{E}_{\mathbb{Q}_t}[S_1] = t\mathbb{E}_{\mathbb{Q}_1}[S_1] + (1-t)\mathbb{E}_{\mathbb{Q}_0}[S_1] = (1+r)S_0$. The corresponding price is $\pi_t = t\pi_1 + (1-t)\pi_0$, so the set of prices is convex. □

So a claim either has a unique no-arbitrage price, or a whole interval of them, in which case the model simply does not determine the price. The good case has a name.

Definition 4.15 (Attainable claim)

A claim Y is **attainable** (or **replicable**) if $Y = a + b^\top S_1$ for some $a \in \mathbb{R}$, $b \in \mathbb{R}^d$; equivalently, if $Y \in \mathcal{X}(y)$ for some y. The portfolio achieving this is called a **replicating portfolio** or a **hedge**.

Theorem 4.16

If $Y \in \mathcal{X}(y)$ is attainable, then its unique no-arbitrage price is y, the initial cost of the replicating portfolio.

Proof. Write $Y = a + b^\top S_1$, so $y = a/(1+r) + b^\top S_0$. For any risk-neutral $\mathbb{Q}$,

$$\frac{1}{1+r}\mathbb{E}_\mathbb{Q}[Y] = \frac{a}{1+r} + \frac{1}{1+r}b^\top\mathbb{E}_\mathbb{Q}[S_1] = \frac{a}{1+r} + b^\top S_0 = y,$$

independently of $\mathbb{Q}$. By Theorem 4.13 this is the unique no-arbitrage price. □

This is the **law of one price**: two portfolios with the same payout must have the same price. Our very first example was an instance – there the call was attainable, $Y = \frac{1}{3}(S_1 - 4)$, and the price $\frac{1}{3}$ was the cost of the hedge.

The converse of Theorem 4.16 is also true, though we shall not prove it here (it is on the example sheet): a claim with a unique no-arbitrage price is attainable.

§4.7 Complete Markets

Definition 4.17 (Complete market)

A market model is **complete** if every contingent claim is attainable.

Proposition 4.18

In a complete market model, S_1 takes at most $d+1$ values.

Proof. Let $A_1, \dots, A_n$ be disjoint events of positive probability. The indicators $\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_n}$ are linearly independent random variables. By completeness each $\mathbb{1}_{A_i} = a_i + b_i^\top S_1$, so

$$\operatorname{span}\{\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_n}\} \subseteq \operatorname{span}\{1, S_1^1, \dots, S_1^d\},$$

a space of dimension at most $d+1$. Hence $n \le d+1$. As S_1 taking n distinct values would give n such disjoint events, the result follows. □

So completeness is a strong requirement: with one stock ($d = 1$) it forces a binomial model. This is precisely why the binomial model is the workhorse of discrete-time finance.

Theorem 4.19 (Second fundamental theorem of asset pricing)

An arbitrage-free market model is complete if and only if the risk-neutral measure is unique.

Proof. ($\Rightarrow$) Suppose the market is complete and $\mathbb{Q}_0, \mathbb{Q}_1$ are both risk-neutral. Set

$$Y = \frac{\mathrm{d}\mathbb{Q}_1}{\mathrm{d}\mathbb{P}} - \frac{\mathrm{d}\mathbb{Q}_0}{\mathrm{d}\mathbb{P}},$$

which is a bounded-below claim (in a complete market S_1 takes finitely many values, so Y is a genuine random variable and everything is integrable). Since Y is attainable, write $Y = a + b^\top S_1$; then for $i = 0, 1$,

$$\mathbb{E}_{\mathbb{Q}_i}[Y] = a + b^\top \mathbb{E}_{\mathbb{Q}_i}[S_1] = a + (1+r)b^\top S_0,$$

the same value for both. But

$$\mathbb{E}_{\mathbb{Q}_1}[Y] - \mathbb{E}_{\mathbb{Q}_0}[Y] = \mathbb{E}_{\mathbb{P}}\left[\left(\frac{\mathrm{d}\mathbb{Q}_1}{\mathrm{d}\mathbb{P}} - \frac{\mathrm{d}\mathbb{Q}_0}{\mathrm{d}\mathbb{P}}\right) Y\right] = \mathbb{E}_{\mathbb{P}}[Y^2],$$

so $\mathbb{E}_{\mathbb{P}}[Y^2] = 0$ and hence $Y = 0$ a.s., i.e. $\mathbb{Q}_0 = \mathbb{Q}_1$.

($\Leftarrow$) If the risk-neutral measure is unique then by Theorem 4.13 every claim has exactly one no-arbitrage price, and hence (by the converse to Theorem 4.16) is attainable. □

Example 4.20

Return to Example 4.7. If $a < r < b$ the model is arbitrage-free with a unique risk-neutral measure $q = (r-a)/(b-a)$, hence complete: every claim $Y = g(S_1)$ is replicable, and

$$\pi = \frac{1}{1+r}\Big(q\,g((1+b)S_0) + (1-q)\,g((1+a)S_0)\Big).$$

The replicating portfolio is obtained by solving two linear equations in two unknowns; the number of shares held is

$$\theta = \frac{g((1+b)S_0) - g((1+a)S_0)}{(b-a)S_0},$$

the ratio of the change in payout to the change in stock price. This quantity will reappear as the *delta* of the claim, and it is the single most important number in practical derivative trading.

§4.8 Utility Indifference Pricing

Theorem 4.13 pins down the price of a claim only when the claim is attainable. When it is not, we must go back to preferences. This subsection is a little more technical and may be skipped on a first reading, but it explains *why* the risk-neutral price is the right answer even in incomplete markets, at least for small trades.

Definition 4.21 (Indifference price)

Fix a utility U and initial wealth x. The **utility indifference price** (or **reservation bid price**) of a claim Y is the largest π with

$$\sup\{\mathbb{E}[U(X+Y)] : X \in \mathcal{X}(x-\pi)\} \geq \sup\{\mathbb{E}[U(X)] : X \in \mathcal{X}(x)\}.$$

In words – the most the agent would pay for Y while remaining at least as happy as if they had not bought it. We write $F_Y(\pi)$ for the left-hand side; note that $X \in \mathcal{X}(x-\pi)$ if and only if $X + (1+r)\pi \in \mathcal{X}(x)$, so

$$F_Y(\pi) = \sup\{\mathbb{E}[U(X - (1+r)\pi + Y)] : X \in \mathcal{X}(x)\},$$

which is decreasing and (given mild regularity) continuous in π, with $F_Y(\mp\infty) = U(\pm\infty)$. Thus $\pi(Y)$ is the unique solution of $F_Y(\pi) = F_0(0)$.

Theorem 4.22

The map $Y \mapsto \pi(Y)$ is concave.

Proof. Fix claims Y_0, Y_1 and $t \in [0,1]$, and set $Y_t = tY_1 + (1-t)Y_0$. Let X_i attain the supremum defining $F_{Y_i}(\pi(Y_i))$ for $i = 0, 1$. Then $tX_1 + (1-t)X_0 \in \mathcal{X}(x - t\pi(Y_1) - (1-t)\pi(Y_0))$ by linearity, and so, using concavity of U,

$$\begin{aligned}
F_{Y_t}(\pi(Y_t)) = F_0(0) &= tF_{Y_1}(\pi(Y_1)) + (1-t)F_{Y_0}(\pi(Y_0)) \\
&= t\,\mathbb{E}[U(X_1+Y_1)] + (1-t)\,\mathbb{E}[U(X_0+Y_0)] \\
&\leq \mathbb{E}\big[U(tX_1 + (1-t)X_0 + Y_t)\big] \\
&\leq F_{Y_t}\big(t\pi(Y_1) + (1-t)\pi(Y_0)\big).
\end{aligned}$$

Since F_{Y_t} is decreasing, $\pi(Y_t) \geq t\pi(Y_1) + (1-t)\pi(Y_0)$. □

Concavity says that an investor values diversification: half of one claim plus half of another is worth at least the average of the two. It also implies that $t \mapsto \pi(tY)/t$ is decreasing, so the limit

$$\pi_0 := \lim_{t \downarrow 0} \frac{\pi(tY)}{t}$$

exists. This is the **marginal** price of the claim – the price per unit for an infinitesimally small trade.

Lemma 4.23 (Supporting hyperplane)
If U is concave and differentiable then $U(y) - U(x) \leq U'(x)(y-x)$ for all x, y.

Proof. For $x < x + \epsilon < y$, concavity of U gives that the chord slopes decrease, so

$$\frac{U(y) - U(x)}{y - x} \leq \frac{U(x+\epsilon) - U(x)}{\epsilon}.$$

Letting $\epsilon \downarrow 0$ gives the claim, and the case $y < x$ is symmetric. □

Lemma 4.24
If X^* maximises $\mathbb{E}[U(X+Y)]$ over $X \in \mathcal{X}(x)$, then $\mathbb{E}[U'(X^*+Y)\,\xi] = 0$; consequently $\mathbb{E}[U'(X^*+Y)(X^*-X)] = 0$ for all $X \in \mathcal{X}(x)$.

Proof. Identical to Theorem 3.11, differentiating in θ. The second claim follows since $X^* - X$ is of the form $\eta^\top \xi$. □

Theorem 4.25 (Marginal pricing)
Let X_0 maximise $\mathbb{E}[U(X)]$ over $X \in \mathcal{X}(x)$. Then

$$\pi_0 = \frac{\mathbb{E}[U'(X_0)Y]}{(1+r)\,\mathbb{E}[U'(X_0)]} = \mathbb{E}[ZY],$$

where Z is the state price density of Theorem 3.13.

Proof. Write $\pi_t = \pi(tY)/t$ and let X_t maximise $\mathbb{E}[U(X+tY)]$ over $X \in \mathcal{X}(x - t\pi_t)$. By definition of the indifference price, $\mathbb{E}[U(X_t + tY)] = \mathbb{E}[U(X_0)]$.

For the upper bound, note that $X_0 - t\pi_t(1+r) \in \mathcal{X}(x - t\pi_t)$, so it is a candidate in the supremum defining X_t, whence

$$\begin{aligned} 0 = \frac{\mathbb{E}[U(X_t + tY) - U(X_0)]}{t} &\geq \frac{\mathbb{E}[U(X_0 - t\pi_t(1+r) + tY) - U(X_0)]}{t} \\ &\xrightarrow[t\downarrow 0]{} \mathbb{E}\big[U'(X_0)(Y - (1+r)\pi_0)\big] \end{aligned}$$

using $\pi_t \downarrow \pi_0$ and dominated convergence. So $\mathbb{E}[U'(X_0)(Y - (1+r)\pi_0)] \leq 0$.

For the lower bound, apply Lemma 4.23 and then Lemma 4.24:

$$\begin{aligned} 0 = \frac{\mathbb{E}[U(X_t + tY) - U(X_0)]}{t} &\le \frac{\mathbb{E}[U'(X_0)(X_t + tY - X_0)]}{t} \\ &= \mathbb{E}\big[U'(X_0)(Y - (1+r)\pi_t)\big] \\ &\quad + \tfrac{1}{t}\mathbb{E}\big[U'(X_0)(X_t + t\pi_t(1+r) - X_0)\big] \\ &= \mathbb{E}\big[U'(X_0)(Y - (1+r)\pi_t)\big], \end{aligned}$$

since $X_t + t\pi_t(1+r) \in \mathcal{X}(x)$ and Lemma 4.24 (with $Y = 0$) makes the last expectation vanish. Letting $t \downarrow 0$ gives $\mathbb{E}[U'(X_0)(Y - (1+r)\pi_0)] \ge 0$.

Combining the two bounds gives equality, which rearranges to the stated formula. □

So for small trades the indifference price is linear, and the linear functional is given by a state price density – equivalently, by a risk-neutral expectation. This is the sense in which risk-neutral pricing is the right answer even when replication is impossible: it is the first-order approximation to what an optimising agent would pay.

§5 Martingales in Discrete Time

To do finance in more than one period we must be able to say what an agent knows at each time. The mathematical device for this is the σ-algebra, and the resulting theory – conditional expectation and martingales – is the technical backbone of everything that follows.

§5.1 Information and σ-Algebras

Suppose we toss a coin twice, so $\Omega = \{HH, HT, TH, TT\}$. After the first toss we know whether the outcome lies in $\{HH, HT\}$ or in $\{TH, TT\}$, but nothing more. So the events we can resolve are

$$\mathcal{G} = \big\{\emptyset,\ \{HH, HT\},\ \{TH, TT\},\ \Omega\big\}.$$

The general shape of such a collection is clear: if we can resolve A we can resolve A^c; if we can resolve each of $A_1, A_2, \dots$ we can resolve $\bigcup_n A_n$; and we can always resolve $\emptyset$. That is exactly the definition of a σ-algebra.

Definition 5.1 (σ-algebra)

A **σ-algebra** on a set Ω is a collection $\mathcal{G}$ of subsets of Ω with $\emptyset \in \mathcal{G}$, closed under complements and under countable unions. A random variable X is **$\mathcal{G}$-measurable** if $X^{-1}(B) \in \mathcal{G}$ for every Borel set $B \subseteq \mathbb{R}$; equivalently, if $\{X \le x\} \in \mathcal{G}$ for every $x \in \mathbb{R}$.

The intuition is: $\mathcal{G}$ is 'the information available', and X is $\mathcal{G}$-measurable if X is known once that information is revealed. The two extreme cases are the **trivial σ-algebra** $\{\emptyset, \Omega\}$ (no information – the only measurable random variables are constants) and the power set 2^Ω (everything is known).

Definition 5.2

The σ-algebra **generated** by a random variable X is $\sigma(X) = \{X^{-1}(B) : B \text{ Borel}\}$, the smallest σ-algebra making X measurable. More generally $\sigma(X_1, \dots, X_n)$ is the smallest σ-algebra making all of $X_1, \dots, X_n$ measurable.

Fact 5.3

A random variable Y is $\sigma(X)$-measurable if and only if $Y = g(X)$ for some Borel measurable g.

This is the precise sense in which 'knowing X' means 'knowing every function of X, and nothing else'. We shall use it without further comment.

§5.2 Conditional Expectation

In Part IA Probability you defined $\mathbb{E}[X \mid A] = \mathbb{E}[X\mathbb{1}_A]/\mathbb{P}(A)$ for an event A of positive probability, and $\mathbb{E}[X \mid Y]$ for a discrete random variable Y as the random variable taking value $\mathbb{E}[X \mid Y = y]$ on $\{Y = y\}$. We now want to condition on a whole σ-algebra, including ones generated by continuous random variables where every individual event has probability zero. The trick is to characterise the answer rather than construct it.

Note first the key property of the discrete construction: if $f(y) = \mathbb{E}[X \mid Y = y]$ then for every bounded $\sigma(Y)$-measurable Z,

$$\mathbb{E}[XZ] = \mathbb{E}[f(Y)Z],$$

as one checks by decomposing over the values of Y. We turn this into a definition.

Definition 5.4 (Conditional expectation)

Let X be an integrable random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G} \subseteq \mathcal{F}$ a σ-algebra. A random variable $\tilde{X}$ is a **conditional expectation** of X given $\mathcal{G}$ if

(i) $\tilde{X}$ is $\mathcal{G}$-measurable and integrable;

(ii) $\mathbb{E}[XZ] = \mathbb{E}[\tilde{X}Z]$ for every bounded $\mathcal{G}$-measurable Z.

Proposition 5.5

A conditional expectation exists and is almost surely unique. We write it $\mathbb{E}[X \mid \mathcal{G}]$, and $\mathbb{E}[X \mid Y] := \mathbb{E}[X \mid \sigma(Y)]$.

Proof. We prove uniqueness, existence being a genuinely measure-theoretic matter (it follows from the Radon–Nikodym theorem, or from the L^2 projection argument below). Suppose $\tilde{X}_0$ and $\tilde{X}_1$ both work. Take $Z = \mathbb{1}_{\{\tilde{X}_0 < \tilde{X}_1\}}$, which is bounded and $\mathcal{G}$-measurable. Then

$$\mathbb{E}\big[(\tilde{X}_1 - \tilde{X}_0)\mathbb{1}_{\{\tilde{X}_0 < \tilde{X}_1\}}\big] = \mathbb{E}[XZ] - \mathbb{E}[XZ] = 0.$$

The integrand is non-negative and strictly positive on $\{\tilde{X}_0 < \tilde{X}_1\}$, so $\mathbb{P}(\tilde{X}_0 < \tilde{X}_1) = 0$. By symmetry $\mathbb{P}(\tilde{X}_0 > \tilde{X}_1) = 0$. □

The following interpretation is worth recording, since it explains why conditional expectation is 'the best guess'.

Proposition 5.6 (L^2 projection)

Let X be square integrable and $\tilde{X} = \mathbb{E}[X \mid \mathcal{G}]$. Then for every square-integrable $\mathcal{G}$-measurable Z,

$$\mathbb{E}\big[(X-\tilde{X})^2\big] \le \mathbb{E}\big[(X-Z)^2\big].$$

Proof. Expand:

$$\mathbb{E}[(X-Z)^2] = \mathbb{E}[(X-\tilde{X})^2] + 2\mathbb{E}\big[(X-\tilde{X})(\tilde{X}-Z)\big] + \mathbb{E}[(\tilde{X}-Z)^2].$$

The middle term vanishes: $\tilde{X} - Z$ is $\mathcal{G}$-measurable, so the defining property gives $\mathbb{E}[X(\tilde{X}-Z)] = \mathbb{E}[\tilde{X}(\tilde{X}-Z)]$ (first for bounded $\tilde{X}-Z$, then in general by truncation). The last term is non-negative. □

So $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace of $\mathcal{G}$-measurable square-integrable random variables. All the familiar properties now follow.

Proposition 5.7 (Properties of conditional expectation)

Assume all the conditional expectations below are well defined. Then, almost surely:

(i) (*Linearity*) $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$;

(ii) (*Taking out what is known*) if Y is $\mathcal{G}$-measurable then $\mathbb{E}[XY \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]$;

(iii) (*Positivity*) if $X \ge 0$ a.s. then $\mathbb{E}[X \mid \mathcal{G}] \ge 0$ a.s.;

(iv) (*Independence*) if $\sigma(X)$ is independent of $\mathcal{G}$ then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$;

(v) (*Jensen*) if f is convex then $\mathbb{E}[f(X) \mid \mathcal{G}] \ge f(\mathbb{E}[X \mid \mathcal{G}])$;

(vi) (*Tower property*) if $\mathcal{G} \subseteq \mathcal{H} \subseteq \mathcal{F}$ then $\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]$.

Proof. (i) Both sides are $\mathcal{G}$-measurable and satisfy the defining identity, so uniqueness applies. (ii) For bounded $\mathcal{G}$-measurable Z we have $\mathbb{E}[(XY)Z] = \mathbb{E}[X(YZ)] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]YZ]$, since YZ is $\mathcal{G}$-measurable (first for bounded Y, then by truncation). (iii) Take $Z = \mathbb{1}_{\{\mathbb{E}[X|\mathcal{G}]<0\}}$; then $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]Z] = \mathbb{E}[XZ] \ge 0$, whereas the left-hand side is ≤ 0 and is < 0 unless $\mathbb{P}(\mathbb{E}[X \mid \mathcal{G}] < 0) = 0$. (iv) Immediate from the definition, since $\mathbb{E}[XZ] = \mathbb{E}[X]\mathbb{E}[Z]$ for $\mathcal{G}$-measurable Z. (v) A convex f satisfies $f(y) \ge f(x) + \lambda(x)(y-x)$ for a suitable $\lambda(x)$; put $x = \mathbb{E}[X \mid \mathcal{G}]$, $y = X$ and condition. (vi) $\mathbb{E}[X \mid \mathcal{G}]$ is $\mathcal{H}$-measurable so equals its own conditional expectation given $\mathcal{H}$; for the other order, any bounded $\mathcal{G}$-measurable Z is $\mathcal{H}$-measurable, so $\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}]Z] = \mathbb{E}[XZ]$. □

Example 5.8

Let $(A_n)_{n\geq 1}$ partition Ω with $\mathbb{P}(A_n) > 0$ and let $\mathcal{G} = \sigma(A_1, A_2, \dots)$. A random variable is $\mathcal{G}$-measurable exactly when it is constant on each A_n, and

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_n \mathbb{E}[X \mid A_n] \, \mathbb{1}_{A_n},$$

which recovers the Part IA definition.

§5.3 Filtrations and Martingales

Definition 5.9 (Filtration)

A **filtration** on $(\Omega, \mathcal{F})$ is an increasing sequence $(\mathcal{F}_n)_{n\geq 0}$ of σ-algebras, $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}$. A stochastic process $(X_n)_{n\geq 0}$ is **adapted** to $(\mathcal{F}_n)$ if X_n is $\mathcal{F}_n$-measurable for every n. The filtration **generated** by a process (X_n) is $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$.

We adopt the convention that $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is trivial, so X_0 is a constant and $\mathbb{E}[Z \mid \mathcal{F}_0] = \mathbb{E}[Z]$ for every integrable Z.

Definition 5.10 (Martingale)

An adapted, integrable process $(X_n)_{n\geq 0}$ is a **martingale** with respect to $(\mathcal{F}_n)$ if

$$\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] = X_{n-1} \qquad \text{for all } n \geq 1.$$

It is a **supermartingale** if $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \leq X_{n-1}$ and a **submartingale** if $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \geq X_{n-1}$.

A martingale is a fair game: given everything you know today, your expected fortune tomorrow is exactly your fortune today. A supermartingale is a game which is unfavourable to you (it goes *down* on average – the terminology is unfortunate but universal), and a submartingale is a favourable one.

Proposition 5.11

For an adapted integrable process (X_n) the following are equivalent:

(i) (X_n) is a martingale;

(ii) $\mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] = 0$ for all $n \geq 1$;

(iii) $\mathbb{E}[X_n \mid \mathcal{F}_m] = X_m$ for all $0 \leq m \leq n$.

Proof. (i) $\iff$ (ii) is immediate from linearity, and (iii) $\implies$ (i) is the case $m = n-1$. For (i) $\implies$ (iii), induct on $n - m$ using the tower property:

$$\mathbb{E}[X_n \mid \mathcal{F}_m] = \mathbb{E}\big[\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \,\big|\, \mathcal{F}_m\big] = \mathbb{E}[X_{n-1} \mid \mathcal{F}_m] = X_m. \qquad \square$$

Condition (iii) makes no reference to 'the previous time step', and so is the definition that generalises to continuous time.

Example 5.12 (Martingales built backwards)

Let Z be integrable and $(\mathcal{F}_n)$ a filtration. Then $X_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ is a martingale. Integrability follows from conditional Jensen, $\mathbb{E}|X_n| \le \mathbb{E}[\mathbb{E}[|Z| \mid \mathcal{F}_n]] = \mathbb{E}|Z| < \infty$, and the martingale property is the tower property. Every martingale on a finite horizon arises this way, with $Z = X_N$.

Example 5.13 (Martingales built forwards)

(i) Let (Y_n) be independent integrable random variables with $\mathbb{E}[Y_n] = 0$, and let $\mathcal{F}_n = \sigma(Y_1, \dots, Y_n)$. Then $S_0 = 0$, $S_n = Y_1 + \cdots + Y_n$ defines a martingale, since

$$\mathbb{E}[S_n - S_{n-1} \mid \mathcal{F}_{n-1}] = \mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] = \mathbb{E}[Y_n] = 0$$

by independence. In particular the simple symmetric random walk is a martingale.

(ii) If instead (Y_n) are independent, positive, with $\mathbb{E}[Y_n] = 1$, then $M_0 = 1$, $M_n = Y_1 \cdots Y_n$ is a martingale by the same computation.

(iii) If (S_n) is a simple symmetric random walk then $Q_n = S_n^2 - n$ is a martingale: indeed $\mathbb{E}[S_n^2 \mid \mathcal{F}_{n-1}] = S_{n-1}^2 + 2S_{n-1}\mathbb{E}[Y_n] + \mathbb{E}[Y_n^2] = S_{n-1}^2 + 1$.

The definition immediately relevant to us is the following, which we shall justify in §6.

Definition 5.14 (Risk-neutral measure, multi-period)

For a multi-period market with prices $(S_n)_{n\ge0}$ adapted to $(\mathcal{F}_n)$ and interest rate r, a probability measure $\mathbb{Q} \sim \mathbb{P}$ is **risk-neutral** if the discounted price process $\big(S_n/(1+r)^n\big)_{n\ge0}$ is a $\mathbb{Q}$-martingale, i.e. if

$$\mathbb{E}_{\mathbb{Q}}[S_n \mid \mathcal{F}_{n-1}] = (1+r)S_{n-1} \qquad \text{for all } n \ge 1.$$

§5.4 Stopping Times and Optional Stopping

Definition 5.15 (Stopping time)

A random variable T taking values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ is a **stopping time** for the filtration $(\mathcal{F}_n)$ if $\{T \le n\} \in \mathcal{F}_n$ for every n.

The condition says that we can decide whether to stop at time n using only information available at time n – no peeking into the future.

Example 5.16

(i) If (X_n) is adapted then the first hitting time $T = \inf\{n \ge 0 : X_n > 0\}$ (with $\inf\emptyset = \infty$) is a stopping time, since $\{T \le n\} = \bigcup_{k\le n}\{X_k > 0\} \in \mathcal{F}_n$.

(ii) The last time $T = \sup\{n \ge 0 : X_n > 0\}$ is generally *not* a stopping time: $\{T \le n\} = \bigcap_{k>n}\{X_k \le 0\}$ requires knowledge of the whole future. ‘Sell at the top’ is not an implementable strategy.

Definition 5.17 (Stopped process)

Given a process X and a random time T, the **stopped process** X^T is defined by

$$X_n^T = X_{n\wedge T} = X_0 + \sum_{k=1}^{n} \mathbb{1}_{\{k\le T\}}(X_k - X_{k-1}).$$

Proposition 5.18

If X is integrable and adapted and T is a stopping time, then X^T is integrable and adapted.

Proof. Integrability follows from the triangle inequality applied to the sum. For adaptedness, note $\{k \le T\} = \{T \le k-1\}^c \in \mathcal{F}_{k-1} \subseteq \mathcal{F}_n$ for $k \le n$, and $X_k - X_{k-1}$ is $\mathcal{F}_n$-measurable. $\square$

Theorem 5.19 (Optional sampling)

Let X be a martingale and T a stopping time. Then X^T is a martingale. In particular, if $T \le N$ almost surely for some constant N, then $\mathbb{E}[X_T] = X_0$.

Proof. We know X^T is adapted and integrable, so it remains to compute, for $n \ge 1$,

$$\begin{aligned}\mathbb{E}\big[X_n^T - X_{n-1}^T \mid \mathcal{F}_{n-1}\big] &= \mathbb{E}\big[\mathbb{1}_{\{n\le T\}}(X_n - X_{n-1}) \mid \mathcal{F}_{n-1}\big] \\ &= \mathbb{1}_{\{n\le T\}}\,\mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] = 0,\end{aligned}$$

where we took out the $\mathcal{F}_{n-1}$-measurable factor $\mathbb{1}_{\{n\le T\}}$. If $T \le N$ a.s. then $X_T = X_{T\wedge N} = X_N^T$, and $\mathbb{E}[X_N^T] = X_0^T = X_0$. $\square$

The boundedness hypothesis is essential.

Example 5.20 (Why boundedness matters)

Let (S_n) be a simple symmetric random walk from 0 and let $T = \inf\{n \ge 0 : S_n = b\}$ for $b > 0$. From Part IB Markov Chains we know the walk is recurrent, so $T < \infty$ a.s. and $S_T = b$. Then $\mathbb{E}[S_T] = b \ne 0 = S_0$. Optional sampling fails, because T is unbounded and unboundedly large excursions occur before T.

This is not a mere technicality: it is the martingale-theoretic reason why the 'double your bet until you win' strategy does not actually work. It requires an unbounded amount of credit, and $\mathbb{E}[\min_{n\le T} S_n] = -\infty$.

To handle unbounded stopping times we need an integrability condition.

Theorem 5.21 (Optional stopping)

Let X be a martingale and T a stopping time with $T < \infty$ a.s. Suppose there is an integrable random variable Y with $|X_n^T| \le Y$ for all n. Then $X_n^T \to X_T$ a.s. and $\mathbb{E}[X_T] = X_0$.

Proof. Since $T < \infty$ a.s., $X_n^T = X_T$ for all $n \geq T$, giving almost sure convergence. By Theorem 5.19, $\mathbb{E}[X_n^T] = X_0$ for every n; the dominated convergence theorem lets us pass to the limit. □

In particular the conclusion holds if T is bounded, or if T is integrable and the increments of X are bounded, or if X^T is bounded. Let us see the last of these in action.

Example 5.22 (Gambler's ruin, again)

Let (S_n) be a simple symmetric random walk from 0 and $T = \inf\{n \geq 0 : S_n \in \{-a, b\}\}$ for integers $a, b > 0$. Then $|S_n^T| \leq \max(a, b)$, and $T < \infty$ a.s. (again by recurrence), so Theorem 5.21 gives

$$0 = \mathbb{E}[S_T] = -a\,\mathbb{P}(S_T = -a) + b\,\mathbb{P}(S_T = b).$$

With $\mathbb{P}(S_T = -a) + \mathbb{P}(S_T = b) = 1$ this yields

$$\mathbb{P}(S_T = -a) = \frac{b}{a+b}, \qquad \mathbb{P}(S_T = b) = \frac{a}{a+b},$$

which is the gambler's ruin formula from Part IA. Applying the same argument to the martingale $Q_n = S_n^2 - n$ and the bounded stopping time $T \wedge n$,

$$\mathbb{E}[S_{T\wedge n}^2] = \mathbb{E}[T \wedge n].$$

The right-hand side increases to $\mathbb{E}[T]$ by monotone convergence and the left-hand side converges to $\mathbb{E}[S_T^2]$ by dominated convergence, so

$$\mathbb{E}[T] = \mathbb{E}[S_T^2] = a^2 \frac{a}{a+b} \cdot \frac{b}{a} + b^2 \frac{a}{a+b} = ab,$$

after simplification: the expected duration is ab.

§5.5 Martingale Transforms

Definition 5.23 (Previsible process)

A process $(H_n)_{n\geq 1}$ is **previsible** (or **predictable**) for $(\mathcal{F}_n)$ if H_n is $\mathcal{F}_{n-1}$-measurable for every $n \geq 1$.

Definition 5.24 (Martingale transform)

The **martingale transform** of X by H is the process

$$(H \cdot X)_n = \sum_{k=1}^{n} H_k (X_k - X_{k-1}), \qquad (H \cdot X)_0 = 0.$$

Think of $X_k - X_{k-1}$ as the winnings per unit stake in game k and H_k as the stake, chosen before the game is played. Then $(H \cdot X)_n$ is the total winnings. The martingale transform is the discrete-time ancestor of the stochastic integral $\int H \,\mathrm{d}X$.

Theorem 5.25 (You cannot beat the system)
Let H be previsible and bounded.

(i) If X is a martingale then so is $H \cdot X$.

(ii) If X is a supermartingale and $H \geq 0$, then $H \cdot X$ is a supermartingale.

Proof. In both cases $H \cdot X$ is adapted (since H_k is $\mathcal{F}_{k-1}$-measurable and X adapted) and integrable (since H is bounded). Then

$$\mathbb{E}\big[(H\cdot X)_n-(H\cdot X)_{n-1} \mid \mathcal{F}_{n-1}\big] = \mathbb{E}\big[H_n(X_n-X_{n-1}) \mid \mathcal{F}_{n-1}\big] = H_n\,\mathbb{E}[X_n-X_{n-1} \mid \mathcal{F}_{n-1}],$$

which is 0 in case (i) and ≤ 0 in case (ii) as $H_n \geq 0$. □

The name is apt: no gambling strategy, however clever, can turn a fair game into a favourable one, provided your stakes are chosen without knowledge of the future and you are not allowed to bet unboundedly.

Notice that the stopped process is a special case: $X^T - X_0 = H \cdot X$ with $H_k = \mathbb{1}_{\{k \leq T\}}$, which is previsible precisely because T is a stopping time.

§5.6 Supermartingales and the Snell Envelope

For optimal stopping problems we need the supermartingale analogue of optional sampling.

Theorem 5.26 (Optional sampling for supermartingales)
Let X be a supermartingale and let $S \leq T$ be bounded stopping times. Then $\mathbb{E}[X_S] \geq \mathbb{E}[X_T]$.

Proof. Suppose $T \leq N$. Writing the difference as a sum,

$$X_{N\wedge T} - X_{N\wedge S} = \sum_{k=1}^{N} \mathbb{1}_{\{S<k\leq T\}}(X_k - X_{k-1}).$$

Now $\{S < k\} = \{S \leq k-1\} \in \mathcal{F}_{k-1}$ and $\{k \leq T\} = \{T \leq k-1\}^c \in \mathcal{F}_{k-1}$, so $\mathbb{1}_{\{S<k\leq T\}}$ is $\mathcal{F}_{k-1}$-measurable. Hence

$$\mathbb{E}[X_T - X_S] = \sum_{k=1}^{N} \mathbb{E}\Big[\mathbb{1}_{\{S<k\leq T\}}\,\mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}]\Big] \leq 0,$$

since each conditional expectation is ≤ 0 and the indicator is non-negative. □

Now consider the following optimal stopping problem. Fix a horizon N, an adapted process $(Y_n)_{0\leq n\leq N}$ of payouts, and ask for

$$\sup\big\{\mathbb{E}[Y_T] : T \text{ a stopping time with } T \leq N\big\}.$$

Definition 5.27 (Snell envelope)
The **Snell envelope** of $(Y_n)_{n\le N}$ is the process defined by backward recursion

$$Z_N = Y_N, \qquad Z_{n-1} = \max\left\{Y_{n-1},\ \mathbb{E}[Z_n \mid \mathcal{F}_{n-1}]\right\}.$$

At each time we compare the payout from stopping now with the expected value of carrying on optimally; the Snell envelope is the value of the problem.

Theorem 5.28 (Optimal stopping)
The Snell envelope (Z_n) is the smallest supermartingale dominating (Y_n). Moreover

$$T^* = \min\{n \ge 0 : Z_n = Y_n\}$$

is an optimal stopping time, and $Z_0 = \mathbb{E}[Y_{T^*}] = \sup_{T\le N} \mathbb{E}[Y_T]$.

Proof. By construction $Z_n \ge Y_n$ and $Z_{n-1} \ge \mathbb{E}[Z_n \mid \mathcal{F}_{n-1}]$, so Z is a supermartingale dominating Y. If W is any other such supermartingale then $W_N \ge Y_N = Z_N$, and inductively

$$W_{n-1} \ge \max\{Y_{n-1}, \mathbb{E}[W_n \mid \mathcal{F}_{n-1}]\} \ge \max\{Y_{n-1}, \mathbb{E}[Z_n \mid \mathcal{F}_{n-1}]\} = Z_{n-1},$$

so Z is the smallest.

For any stopping time $T \le N$, Theorem 5.26 applied with $S = 0$ gives $\mathbb{E}[Y_T] \le \mathbb{E}[Z_T] \le Z_0$.

It remains to show equality for T^*. The definition of T^* gives $Z_n > Y_n$ for $n < T^*$, so on $\{n \le T^*\}$ we have $Z_{n-1} = \mathbb{E}[Z_n \mid \mathcal{F}_{n-1}]$ for $n \le T^*$; hence the stopped process Z^{T^*} satisfies

$$\mathbb{E}[Z_n^{T^*} - Z_{n-1}^{T^*} \mid \mathcal{F}_{n-1}] = \mathbb{1}_{\{n\le T^*\}}\big(\mathbb{E}[Z_n \mid \mathcal{F}_{n-1}] - Z_{n-1}\big) = 0,$$

so Z^{T^*} is a martingale. Therefore $Z_0 = \mathbb{E}[Z_N^{T^*}] = \mathbb{E}[Z_{T^*}] = \mathbb{E}[Y_{T^*}]$, using $Z_{T^*} = Y_{T^*}$. □

This is the engine that will price American options. Note the structure of the answer: one solves a backward recursion, and the optimal rule is 'stop as soon as the payout from stopping equals the value of the problem'.

§5.7 Convergence

We record, without proof, the two convergence results that we shall occasionally need. The first is the martingale convergence theorem; you will meet it properly in Part II Probability and Measure.

Theorem 5.29 (Martingale convergence)
Let (X_n) be a martingale bounded in L^1, i.e. $\sup_n \mathbb{E}|X_n| < \infty$. Then $X_n \to X_\infty$ almost surely for some integrable X_∞.

In general $\mathbb{E}[X_\infty] \ne X_0$: Example 5.20 is a counterexample if one truncates suitably.

What is needed for the limit to be 'honest' is uniform integrability, and then $X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n]$ for all n.

Theorem 5.30 (Monotone and dominated convergence)
If $0 \le X_n \le X_{n+1}$ a.s. then $\mathbb{E}[X_n] \to \mathbb{E}[\lim_n X_n]$. If $X_n \to X$ a.s. and $|X_n| \le Y$ for an integrable Y, then $\mathbb{E}[X_n] \to \mathbb{E}[X]$.

§6 Multi-Period Models

We now let the market run for N periods. The state space carries a filtration $(\mathcal{F}_n)_{0 \le n \le N}$, prices (S_n) are adapted, and the bank account grows by a factor $1 + r$ each period. Everything from §4 will survive, with 'random variable' replaced by 'adapted process' and 'expectation' by 'conditional expectation'.

§6.1 Self-Financing Portfolios

Let $\theta_n \in \mathbb{R}^d$ be the portfolio of risky assets held over the interval $(n-1, n]$, and θ_n^0 the amount in the bank over the same interval. These must be chosen at time $n-1$, so we require $(\theta_n)_{n \ge 1}$ to be *previsible*. Let X_n be the agent's wealth at time n.

Definition 6.1 (Self-financing)
A previsible portfolio process (θ_n) is **self-financing** if no money is added to or removed from the portfolio after time 0; that is, if

$$X_{n-1} = \theta_n^0 + \theta_n^\top S_{n-1} \qquad \text{and} \qquad X_n = \theta_n^0(1+r) + \theta_n^\top S_n$$

for all $n \ge 1$.

Eliminating θ_n^0 exactly as in the one-period case gives the wealth recursion

$$X_n = (1+r)X_{n-1} + \theta_n^\top\big(S_n - (1+r)S_{n-1}\big).$$

It is neater in discounted terms. Write $\tilde{X}_n = X_n/(1+r)^n$ and $\tilde{S}_n = S_n/(1+r)^n$ for the discounted wealth and prices, i.e. prices measured in units of the bank account rather than in pounds. Dividing the recursion by $(1+r)^n$,

$$\tilde{X}_n - \tilde{X}_{n-1} = \theta_n^\top\big(\tilde{S}_n - \tilde{S}_{n-1}\big),$$

and summing,

$$\boxed{\tilde{X}_n = X_0 + \sum_{k=1}^n \theta_k^\top\big(\tilde{S}_k - \tilde{S}_{k-1}\big) = X_0 + (\theta \cdot \tilde{S})_n.}$$

In other words – the discounted wealth of a self-financing strategy is the initial wealth plus the martingale transform of the strategy against the discounted prices. Theorem 5.25 now does an enormous amount of work for us.

Remark. The change from pounds to units of the bank account is our first **change of numeraire**. A **numeraire** is any asset with strictly positive price which we use as the unit of account; discounting is the statement that in the numeraire of the bank account the riskless asset has constant price 1. Other choices are possible and sometimes much more convenient; we take this up in §6.3.

§6.2 Arbitrage in Multi-Period Models

Definition 6.2 (Arbitrage)

A previsible process $(\theta_n)_{1\le n\le N}$ is an **arbitrage** if the terminal discounted gain $G = \sum_{k=1}^N \theta_k^\top(\tilde{S}_k - \tilde{S}_{k-1})$ satisfies

$$\mathbb{P}(G \ge 0) = 1 \qquad \text{and} \qquad \mathbb{P}(G > 0) > 0.$$

Theorem 6.3 (Fundamental theorem of asset pricing)

A finite-horizon market model admits no arbitrage if and only if there exists a risk-neutral measure, that is, a $\mathbb{Q} \sim \mathbb{P}$ under which $(\tilde{S}_n)_{0\le n\le N}$ is a martingale.

Proof. ($\Leftarrow$) Let $\mathbb{Q}$ be risk-neutral and let θ be previsible and bounded. By Theorem 5.25 the discounted wealth $\tilde{X}_n = X_0 + (\theta \cdot \tilde{S})_n$ is a $\mathbb{Q}$-martingale, so

$$\mathbb{E}_{\mathbb{Q}}[G] = \mathbb{E}_{\mathbb{Q}}[\tilde{X}_N] - X_0 = 0.$$

If θ were an arbitrage then $G \ge 0$ $\mathbb{P}$-a.s., hence $\mathbb{Q}$-a.s. since $\mathbb{P} \sim \mathbb{Q}$; a non-negative random variable of zero mean vanishes a.s., so $G = 0$ $\mathbb{Q}$-a.s. and therefore $\mathbb{P}$-a.s., contradicting $\mathbb{P}(G > 0) > 0$. (For unbounded θ one first localises using the stopping times $T_m = \inf\{n : \|\theta_{n+1}\| > m\}$ and then lets $m \to \infty$.)

($\Rightarrow$) Conversely, suppose there is no arbitrage. Then in particular, for each n there is no one-period arbitrage conditional on $\mathcal{F}_{n-1}$: if η were an $\mathcal{F}_{n-1}$-measurable random vector with $\eta^\top(\tilde{S}_n - \tilde{S}_{n-1}) \ge 0$ a.s. and positive with positive probability, then the strategy which is η at time n and zero otherwise would be an arbitrage. Applying the one-period Theorem 4.10 conditionally on $\mathcal{F}_{n-1}$ produces, for each n, a strictly positive $\mathcal{F}_n$-measurable Y_n with $\mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] = 1$ and $\mathbb{E}[Y_n\tilde{S}_n \mid \mathcal{F}_{n-1}] = \tilde{S}_{n-1}$. Setting $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = \prod_{n=1}^N Y_n$ defines an equivalent measure, and a short computation with the tower property shows $\tilde{S}$ is a $\mathbb{Q}$-martingale. □

Theorem 6.4 (Second fundamental theorem)

An arbitrage-free model is complete – every $\mathcal{F}_N$-measurable claim Y is replicable by a self-financing strategy – if and only if the risk-neutral measure is unique. In that case the unique time-n no-arbitrage price of a claim Y payable at time N is

$$\pi_n = \frac{1}{(1+r)^{N-n}}\,\mathbb{E}_{\mathbb{Q}}\big[Y \mid \mathcal{F}_n\big].$$

Proof sketch. The equivalence is proved exactly as in Theorem 4.19. For the formula, note that $(\pi_n/(1+r)^n)$ must be a $\mathbb{Q}$-martingale (else the augmented market has an arbitrage, by the argument of Theorem 4.13 applied at each time), with terminal value $Y/(1+r)^N$; and a martingale is determined by its terminal value via Proposition 5.11(iii). □

§6.3 Change of Numeraire

We took it for granted above that the natural thing to divide by is the bank account. It is not the only choice, and being flexible about it is a genuinely useful technique.

Definition 6.5 (Numeraire)

A **numeraire** is an adapted process $(M_n)_{0\le n\le N}$ with $M_n > 0$ for all n, which is the price process of a self-financing portfolio. Prices measured in units of the numeraire are S_n/M_n.

The bank account $M_n = (1+r)^n$ is the numeraire we have used so far; another asset, or the price of a bond maturing at a fixed date, will do just as well. The point is that arbitrage is a numeraire-independent notion – multiplying every price by a strictly positive process changes neither the sign of a gain nor its vanishing – so the fundamental theorem must hold for each choice, and we may pick whichever makes a given computation easiest.

Theorem 6.6 (Change of numeraire)

Let the market be arbitrage-free with risk-neutral measure $\mathbb{Q}$ (for the bank-account numeraire), and let M be a numeraire with $M_0 = 1$. Define $\mathbb{Q}^M \sim \mathbb{Q}$ by

$$\frac{\mathrm{d}\mathbb{Q}^M}{\mathrm{d}\mathbb{Q}} = \frac{M_N}{(1+r)^N}.$$

Then $\mathbb{Q}^M$ is a probability measure, and (S_n/M_n) is a $\mathbb{Q}^M$-martingale. Moreover the time-n price of a claim Y payable at time N is

$$\pi_n = M_n\,\mathbb{E}_{\mathbb{Q}^M}\left[\frac{Y}{M_N}\,\middle|\,\mathcal{F}_n\right].$$

Proof. Since M is the value of a self-financing portfolio, $\tilde{M}_n = M_n/(1+r)^n$ is a $\mathbb{Q}$-martingale with $\tilde{M}_0 = 1$, so the proposed density is positive with $\mathbb{Q}$-expectation 1 and $\mathbb{Q}^M$ is an equivalent probability measure. For a positive $\mathcal{F}_m$-measurable random variable Z the density relation together with the tower property gives the conditional Bayes formula

$$\mathbb{E}_{\mathbb{Q}^M}[Z \mid \mathcal{F}_n] = \frac{\mathbb{E}_{\mathbb{Q}}\left[Z\,\tilde{M}_m \mid \mathcal{F}_n\right]}{\tilde{M}_n}, \qquad n \le m,$$

because $\mathbb{E}_{\mathbb{Q}}[\,\mathrm{d}\mathbb{Q}^M/\,\mathrm{d}\mathbb{Q} \mid \mathcal{F}_m] = \tilde{M}_m$. Taking $Z = S_m/M_m$ and $m = N$,

$$\mathbb{E}_{\mathbb{Q}^M}\left[\frac{S_m}{M_m}\,\middle|\,\mathcal{F}_n\right] = \frac{\mathbb{E}_{\mathbb{Q}}[\tilde{S}_m \mid \mathcal{F}_n]}{\tilde{M}_n} = \frac{\tilde{S}_n}{\tilde{M}_n} = \frac{S_n}{M_n},$$

using that $\tilde{S}$ is a $\mathbb{Q}$-martingale. The pricing formula is the same computation applied to π_m/M_m, whose $\mathbb{Q}$-discounted version is a martingale by the second fundamental theorem. $\square$

The formula is worth reading aloud: *the price of anything, measured in units of the numeraire, is a martingale under the measure associated with that numeraire.* We shall use it in §9.2, where the numeraire is a foreign bank account and the change of measure resolves what looks like a paradox about exchange rates.

§6.4 The Binomial Model

We now specialise to the model in which every computation can be carried out explicitly.

Definition 6.7 (Cox–Ross–Rubinstein model)
Take $d = 1$ and let $(\xi_n)_{n\geq 1}$ be i.i.d. with

$$\mathbb{P}(\xi = 1 + b) = p = 1 - \mathbb{P}(\xi = 1 + a), \qquad a < b,\ p \in (0, 1),$$

and set $S_n = S_{n-1}\xi_n = S_0\xi_1 \cdots \xi_n$, with $(\mathcal{F}_n)$ the filtration generated by (S_n). This is the **binomial** or **Cox–Ross–Rubinstein** model.

Proposition 6.8
The binomial model is arbitrage-free if and only if $a < r < b$, and in that case the risk-neutral measure is unique: under $\mathbb{Q}$ the ξ_n are i.i.d. with

$$\mathbb{Q}(\xi = 1 + b) = q = \frac{r - a}{b - a}.$$

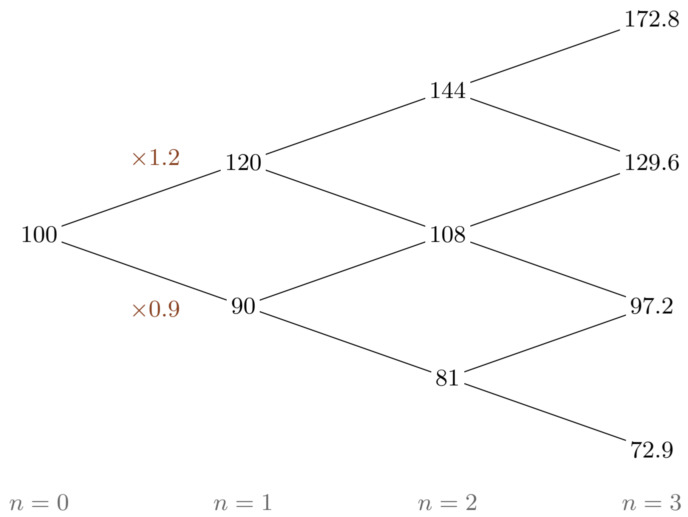
Consequently the model is complete.

Proof. If $r \leq a$ then $\theta_1 = 1$ (buy a share, hold it one period, do nothing thereafter) is an arbitrage, since $S_1 - (1+r)S_0 \geq (1+a-1-r)S_0 \geq 0$ with strict inequality when $\xi_1 = 1 + b$. If $r \geq b$ then $\theta_1 = -1$ is an arbitrage. Conversely, suppose $a < r < b$. A measure $\mathbb{Q}$ under which $\tilde{S}$ is a martingale must satisfy $\mathbb{E}_{\mathbb{Q}}[\xi_n \mid \mathcal{F}_{n-1}] = 1 + r$; since ξ_n takes two values this determines the conditional law of ξ_n given $\mathcal{F}_{n-1}$ to be the two-point law with mass $q = (r - a)/(b - a)$ at $1 + b$, and $q \in (0, 1)$ exactly when $a < r < b$. This makes the ξ_n i.i.d. under $\mathbb{Q}$, and conversely that choice visibly works. Uniqueness plus the second fundamental theorem gives completeness. □

The price process is **recombining**: an up-move followed by a down-move gives the same price as a down followed by an up, so after n steps there are only $n + 1$ possible prices rather than 2^n. Here is the tree we shall use as a running example, with

$$S_0 = 100, \quad 1 + b = 1.2, \quad 1 + a = 0.9, \quad r = 0.05, \quad N = 3,$$

so that $q = (0.05 + 0.1)/(0.2 + 0.1) = \frac{1}{2}$.



§6.5 European Claims: Pricing and Replication

Definition 6.9

A **European contingent claim** with expiry N is an $\mathcal{F}_N$-measurable payout Y, payable at time N. It is **vanilla** if $Y = g(S_N)$.

In the binomial model (S_n) is a Markov chain under $\mathbb{Q}$, so the conditional expectation defining the price depends on $\mathcal{F}_n$ only through S_n. This gives the following.

Theorem 6.10

In an arbitrage-free binomial model, the unique time-n price of the vanilla claim $Y = g(S_N)$ is $\pi_n = V(n, S_n)$, where

$$\begin{aligned} V(n,s) &= \frac{1}{(1+r)^{N-n}} \mathbb{E}_{\mathbb{Q}}\big[g(S_N) \,\big|\, S_n = s\big] \\ &= \frac{1}{(1+r)^{N-n}} \sum_{j=0}^{N-n} \binom{N-n}{j} q^j (1-q)^{N-n-j}\, g\big(s(1+b)^j(1+a)^{N-n-j}\big). \end{aligned}$$

Equivalently, V satisfies the backward recursion

$$(1+r)V(n-1,s) = qV(n, s(1+b)) + (1-q)V(n, s(1+a)), \qquad V(N,s) = g(s).$$

Proof. Under $\mathbb{Q}$ we have $S_N = S_n \xi_{n+1} \cdots \xi_N$ with the ξ's i.i.d. and independent of $\mathcal{F}_n$, so $\mathbb{E}_{\mathbb{Q}}[g(S_N) \mid \mathcal{F}_n]$ is the stated function of S_n, and the number of up-moves in $N-n$ steps is binomial with parameters $N-n$ and q. The recursion is the tower property applied one step at a time. □

The recursion is the practical algorithm: fill in the payouts at expiry, then work backwards through the tree, at each node taking the q-weighted average of the two values to the right and discounting by $1+r$. But pricing is only half the story; a bank that sells an option wants to know how to *hedge* it.

Theorem 6.11 (Replication)

With V as above, define the previsible process

$$\theta_n = \frac{V\big(n, S_{n-1}(1+b)\big) - V\big(n, S_{n-1}(1+a)\big)}{S_{n-1}(b-a)},$$

and let $X_0 = V(0, S_0)$, $X_n = (1+r)X_{n-1} + \theta_n(S_n - (1+r)S_{n-1})$. Then $X_n = V(n, S_n)$ for all n; in particular $X_N = g(S_N)$.

Proof. We induct on n, the case $n = 0$ holding by definition. Suppose $X_{n-1} = V(n-1, S_{n-1})$ and write $V_b = V(n, S_{n-1}(1+b))$, $V_a = V(n, S_{n-1}(1+a))$. Using the recursion of Theorem 6.10,

$$X_n = qV_b + (1-q)V_a + \frac{V_b - V_a}{b-a}(\xi_n - 1 - r).$$

If $\xi_n = 1 + b$ then $\xi_n - 1 - r = b - r$, and since $q = \frac{r-a}{b-a}$, $1 - q = \frac{b-r}{b-a}$,

$$\begin{aligned} X_n &= \frac{(r-a)V_b + (b-r)V_a}{b-a} + \frac{(b-r)(V_b - V_a)}{b-a} \\ &= \frac{(r-a+b-r)V_b + (b-r-b+r)V_a}{b-a} = V_b. \end{aligned}$$

If $\xi_n = 1 + a$ then $\xi_n - 1 - r = a - r$ and the same computation gives $X_n = V_a$. In both cases $X_n = V(n, S_n)$. □

The quantity θ_n is the **delta** of the claim: the ratio of the change in the option value to the change in the stock price over one step. Holding θ_n shares (funded by the bank) exactly reproduces the option, which is what 'hedging' means.

Example 6.12 (A European call in the binomial tree)

Take the tree above and a European call with strike $K = 100$ and expiry $N = 3$. At expiry the payouts are

$$(172.8-100)^+ = 72.8, \quad (129.6-100)^+ = 29.6, \quad (97.2-100)^+ = 0, \quad (72.9-100)^+ = 0.$$

Working backwards with $q = \frac{1}{2}$ and $1 + r = 1.05$:

$$\begin{aligned} V(2, 144) &= \tfrac{1}{1.05}\big(\tfrac{1}{2} \cdot 72.8 + \tfrac{1}{2} \cdot 29.6\big) = \tfrac{51.2}{1.05} = 48.762, \\ V(2, 108) &= \tfrac{1}{1.05}\big(\tfrac{1}{2} \cdot 29.6 + \tfrac{1}{2} \cdot 0\big) = \tfrac{14.8}{1.05} = 14.095, \\ V(2, 81) &= 0, \\ V(1, 120) &= \tfrac{1}{1.05} \cdot \tfrac{1}{2}(48.762 + 14.095) = 29.932, \\ V(1, 90) &= \tfrac{1}{1.05} \cdot \tfrac{1}{2}(14.095 + 0) = 6.712, \\ V(0, 100) &= \tfrac{1}{1.05} \cdot \tfrac{1}{2}(29.932 + 6.712) = 17.450. \end{aligned}$$

As a check, the direct formula of Theorem 6.10 gives

$$V(0, 100) = \frac{1}{1.05^3}\left(\tfrac{1}{8} \cdot 72.8 + \tfrac{3}{8} \cdot 29.6\right) = \frac{9.1 + 11.1}{1.157625} = \frac{20.2}{1.157625} = 17.450.\checkmark$$

The initial hedge is

$$\theta_1 = \frac{29.932 - 6.712}{100 \cdot 0.3} = \frac{23.220}{30} = 0.774,$$

so on selling the option for £17.45 one immediately buys 0.774 shares, costing £77.40, borrowing the difference £59.95 from the bank. One period later the position is rebalanced: for instance if the stock rises to 120, the new delta is $(48.762 - 14.095)/(120 \cdot 0.3) = 0.963$, so one buys a further 0.189 shares.

§6.6 Put–Call Parity

Calls and puts are not independent objects.

Proposition 6.13 (Put–call parity)

Let C_n and P_n be the time-n prices of a European call and put with the same strike K and expiry N. Then

$$C_n - P_n = S_n - \frac{K}{(1+r)^{N-n}}.$$

Proof. At expiry, for every value of S_N,

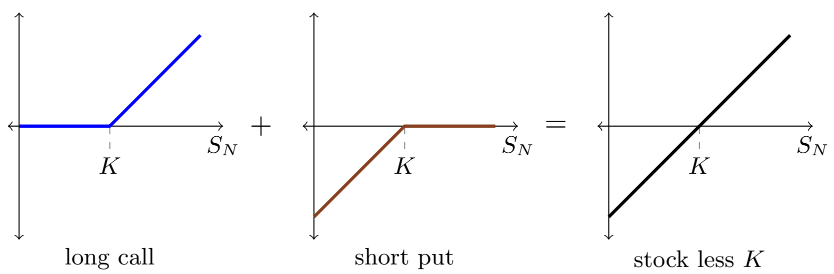
$$(S_N - K)^+ - (K - S_N)^+ = S_N - K,$$

since exactly one of the two options is in the money. Hence a long call and a short put together replicate one share plus a debt of K, and by the law of one price the values must agree. Formally, taking discounted risk-neutral conditional expectations,

$$C_n - P_n = \frac{\mathbb{E}_{\mathbb{Q}}[S_N - K \mid \mathcal{F}_n]}{(1+r)^{N-n}} = S_n - \frac{K}{(1+r)^{N-n}},$$

using that $\tilde{S}$ is a $\mathbb{Q}$-martingale. □

The identity is best seen as a picture: stacking the call payoff on top of the negated put payoff gives the straight line $S_N - K$.



Parity is used constantly in practice: it means that only one of the call and the put need be modelled, and any violation of it is a pure arbitrage requiring no view on the future at all.

Proposition 6.14 (Properties of the call price)

Let $g(s) = (s-K)^+$ and let θ_n be the replicating strategy of Theorem 6.11. Then $s \mapsto V(n,s)$ is increasing and convex, and $\theta_n \in [0,1]$.

Proof. Writing $S_N = S_n Z$ with $Z = \xi_{n+1}\cdots\xi_N$ independent of $\mathcal{F}_n$ and positive,

$$V(n,s) = \frac{1}{(1+r)^{N-n}}\mathbb{E}_{\mathbb{Q}}\big[(sZ-K)^+\big].$$

For each fixed value of Z, the map $s \mapsto (sZ-K)^+$ is increasing and convex; both properties are preserved by taking expectations, so $V(n,\cdot)$ is increasing and convex. Since θ_n is a difference quotient of $V(n,\cdot)$, we get $\theta_n \geq 0$.

For the upper bound use parity in the form $(s-K)^+ = s-K+(K-s)^+$:

$$V(n,s) = s - \frac{K}{(1+r)^{N-n}} + \frac{1}{(1+r)^{N-n}}\mathbb{E}_{\mathbb{Q}}\big[(K-sZ)^+\big],$$

using $\mathbb{E}_{\mathbb{Q}}[Z] = (1+r)^{N-n}$. The final term is decreasing in s, so $s \mapsto V(n,s) - s$ is decreasing, giving $\theta_n \leq 1$. □

The bounds $0 \leq \theta_n \leq 1$ have a clean interpretation: to hedge one call you never need to short the stock, and never need more than one share.

§6.7 American Claims

A European claim may only be exercised at expiry. An American claim may be exercised at any time up to expiry, at the holder's choice.

Definition 6.15 (American claim)

An **American contingent claim** with expiry N is an adapted process $(Y_n)_{0\leq n\leq N}$, where Y_n is the payout if the claim is exercised at time n. The **American call** and **American put** with strike K are $Y_n = (S_n - K)^+$ and $Y_n = (K-S_n)^+$ respectively.

The holder must choose when to exercise using only information available at that time, i.e. must choose a stopping time. So the value is an optimal stopping problem, and §5.6 tells us the answer.

Theorem 6.16

In a complete arbitrage-free market with risk-neutral measure $\mathbb{Q}$, the time-n price of the American claim (Y_n) is

$$\pi_n = \max_{\substack{T \text{ stopping time} \\ n\leq T\leq N}} \mathbb{E}_{\mathbb{Q}}\left[\frac{(1+r)^n}{(1+r)^T}Y_T \;\middle|\; \mathcal{F}_n\right],$$

so that $\pi_n/(1+r)^n$ is the Snell envelope of $Y_n/(1+r)^n$ under $\mathbb{Q}$. Equivalently,

$$\pi_N = Y_N, \qquad \pi_{n-1} = \max\left\{Y_{n-1},\ \frac{1}{1+r}\mathbb{E}_{\mathbb{Q}}[\pi_n \mid \mathcal{F}_{n-1}]\right\},$$

and $T^* = \min\{n : \pi_n = Y_n\}$ is an optimal exercise time.

Proof. This is Theorem 5.28 applied to the discounted payout process under $\mathbb{Q}$, together with the observation that the seller of the claim must be able to hedge against *every* exercise policy, and the writer's minimal hedging cost is exactly the smallest supermartingale dominating the discounted payouts. □

Example 6.17 (An American put)

Take the same tree, and an American put with $K = 100$, $N = 3$. The exercise values are $(100 - S_n)^+$, and we work backwards using $\pi_{n-1} = \max\{Y_{n-1}, \frac{1}{1.05} \cdot \frac{1}{2}(\pi_n^\uparrow + \pi_n^\downarrow)\}$.

At $n = 3$ the payouts are $0,\ 0,\ 2.8,\ 27.1$. At $n = 2$,

$$\begin{aligned} \pi(144) &= \max\{0,\ 0\} = 0, \\ \pi(108) &= \max\left\{0,\ 1.4/1.05\right\} = 1.333, \\ \pi(81) &= \max\left\{19,\ 14.95/1.05\right\} = \max\{19,\ 14.238\} = 19, \end{aligned}$$

so at the node $S_2 = 81$ it is optimal to exercise. At $n = 1$,

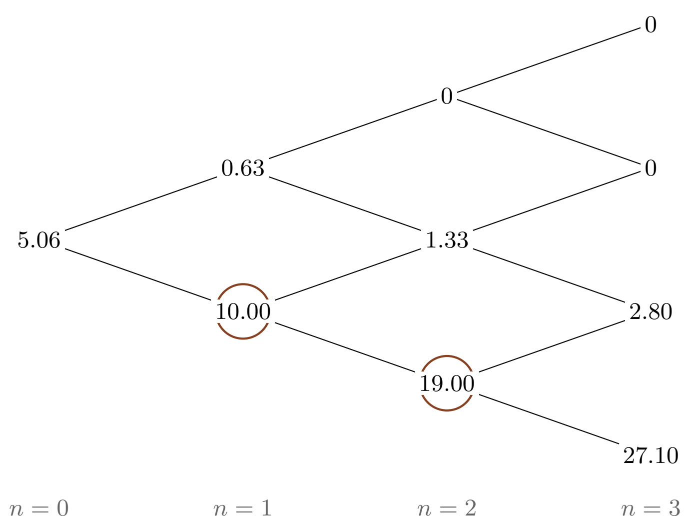
$$\begin{aligned} \pi(120) &= \max\left\{0,\ 0.667/1.05\right\} = 0.635, \\ \pi(90) &= \max\left\{10,\ 10.167/1.05\right\} = \max\{10,\ 9.683\} = 10, \end{aligned}$$

so we exercise at $S_1 = 90$ too. Finally $\pi_0 = \max\{0,\ \frac{1}{1.05} \cdot \frac{1}{2}(0.635 + 10)\} = 5.064$.

For comparison, put–call parity gives the *European* put price

$$P_0 = C_0 + \frac{K}{(1+r)^3} - S_0 = 17.450 + 86.384 - 100 = 3.834,$$

so the right of early exercise is worth $5.064 - 3.834 = 1.230$, a premium of over 30%. The tree of American values, with the exercise nodes ringed, is:



ringed nodes: exercise is optimal

The pattern is typical: one exercises the put when the stock has fallen far enough that the interest earned on the strike outweighs the option value of waiting. For calls the situation is quite different.

Proposition 6.18

Suppose $(Y_n/(1+r)^n)$ is a $\mathbb{Q}$-submartingale. Then it is optimal never to exercise the American claim before expiry, and its price coincides with that of the corresponding European claim.

Proof. We show by backward induction that $\pi_n = \mathbb{E}_{\mathbb{Q}}[Y_N/(1+r)^{N-n} \mid \mathcal{F}_n]$. This holds at $n = N$. If it holds at n then

$$\pi_{n-1} = \max\left\{Y_{n-1}, \ \frac{1}{(1+r)^{N-n+1}}\mathbb{E}_{\mathbb{Q}}[Y_N \mid \mathcal{F}_{n-1}]\right\},$$

and the submartingale property gives $Y_{n-1}/(1+r)^{n-1} \le \mathbb{E}_{\mathbb{Q}}[Y_N/(1+r)^N \mid \mathcal{F}_{n-1}]$, so the second term is the larger. Hence the maximum is attained by never stopping. $\square$

Corollary 6.19 (Never exercise an American call early)

In an arbitrage-free model with $r \ge 0$ and no dividends, an American call has the same value as the European call, and it is never strictly optimal to exercise it early.

Proof. We check that $(S_n - K)^+/(1+r)^n$ is a $\mathbb{Q}$-submartingale. The function $x \mapsto x^+$ is convex, so conditional Jensen gives

$$\mathbb{E}_{\mathbb{Q}}\left[\frac{(S_n-K)^+}{(1+r)^n} \;\middle|\; \mathcal{F}_{n-1}\right] \ge \left(\mathbb{E}_{\mathbb{Q}}\left[\frac{S_n-K}{(1+r)^n} \;\middle|\; \mathcal{F}_{n-1}\right]\right)^+ = \left(\frac{S_{n-1}}{(1+r)^{n-1}} - \frac{K}{(1+r)^n}\right)^+,$$

using that $\tilde{S}$ is a $\mathbb{Q}$-martingale. Since $r \ge 0$ this is at least $\big(S_{n-1} - K\big)^+/(1+r)^{n-1}$, which is the submartingale property. Now apply Proposition 6.18. $\square$

The intuition is worth having: exercising a call early means paying K sooner than necessary and throwing away the insurance that the option provides against the price falling below K. One should sell the option rather than exercise it.

§6.8 Dynamic Programming

The Snell envelope is a special case of a much more general technique, which also solves the multi-period utility maximisation problem. We record it briefly.

Let (ξ_n) be i.i.d., let (U_n) be a previsible **control** process taking values in a set $\mathcal{U}$, and let the **controlled Markov process** (X_n) evolve by

$$X_n = G(X_{n-1}, U_n, \xi_n).$$

We wish to maximise $\mathbb{E}\big[\sum_{k=1}^N f(k, U_k) + g(X_N) \;\big|\; X_0 = x\big]$ over controls.

Definition 6.20

The **value function** is

$$V(n,x) = \sup_{(U_k)} \mathbb{E}\left[\sum_{k=n+1}^{N} f(k,U_k) + g(X_N) \;\middle|\; X_n = x \right].$$

Theorem 6.21 (Bellman)
Suppose V solves **Bellman's equation**

$$V(n-1,x) = \sup_{u\in\mathcal{U}} \Big(f(n,u) + \mathbb{E}\big[V(n, G(x,u,\xi))\big] \Big), \qquad V(N,x) = g(x),$$

and let $u^*(n,x)$ attain the supremum. Then the Markov control $U_n^* = u^*(n, X_{n-1}^*)$ is optimal, and V is the value function.

Proof. Given any control (U_n) with controlled process (X_n), set $M_n = \sum_{k=1}^n f(k,U_k) + V(n,X_n)$. Then

$$\mathbb{E}[M_n - M_{n-1} \mid \mathcal{F}_{n-1}] = f(n,U_n) + \mathbb{E}[V(n,X_n) \mid \mathcal{F}_{n-1}] - V(n-1,X_{n-1}) \le 0$$

by Bellman's equation, so M is a supermartingale, with equality throughout when $U = U^*$, in which case M is a martingale. Hence

$$\mathbb{E}\left[\sum_{k=1}^{N} f(k,U_k) + g(X_N) \right] = \mathbb{E}[M_N] \le M_0 = V(0,x),$$

with equality for $U = U^*$. $\square$

Example 6.22 (Merton's problem with CARA utility)
Take $d=1$, $S_n = S_{n-1}\xi_n$ with (ξ_n) i.i.d., and wealth dynamics $X_n = (1+r)X_{n-1} + \eta_n(\xi_n - (1+r))$, where $\eta_n = \theta_n S_{n-1}$ is the amount of money in the stock – a controlled Markov process. We maximise $\mathbb{E}[U(X_N)]$ for $U(x) = -e^{-\gamma x}$.

Guess $V(n,x) = U\big(x(1+r)^{N-n}\big)A_n$ with $A_N = 1$. Then

$$\begin{aligned} V(n-1,x) &= \sup_{\eta} \mathbb{E}\Big[A_n U\big(((1+r)x + \eta(\xi - (1+r)))(1+r)^{N-n}\big)\Big] \\ &= U\big(x(1+r)^{N-n+1}\big) A_n \inf_t \mathbb{E}\big[e^{t(\xi-(1+r))}\big], \end{aligned}$$

on substituting $t = \gamma\eta(1+r)^{N-n}$. So the guess is consistent with $A_{n-1} = \alpha A_n$, where $\alpha = \inf_t \mathbb{E}[e^{t(\xi-(1+r))}]$, whence $A_n = \alpha^{N-n}$, and the optimal holding is

$$\theta_n^* = \frac{t^*}{\gamma(1+r)^{N-n}S_{n-1}},$$

with t^* the minimiser. Note the CARA signature: the optimal *amount of money* η_n^* invested in the stock is a deterministic constant, independent of current wealth.

Example 6.23 (Consumption and CRRA utility)

If the agent may also consume an amount C_n during $(n-1, n]$, the dynamics become

$$X_n = (1+r)(X_{n-1} - C_n) + \eta_n(\xi_n - (1+r)),$$

and one maximises $\mathbb{E}\big[\sum_{k=1}^N U(C_k) + U(X_N)\big]$. For $U(x) = x^{1-R}/(1-R)$ with $R > 0$, $R \neq 1$, the value function separates as $V(n,x) = U(x)A_n$, and Bellman's equation reduces to a one-dimensional problem in the consumption fraction $s = C/x$:

$$A_{n-1} = \sup_{s\in(0,1)} \left(s^{1-R} + A_n\alpha(1-s)^{1-R}\right), \qquad \alpha = (1-R)\sup_t \mathbb{E}\big[U\big((1+r)+t(\xi-(1+r))\big)\big].$$

Solving gives $A_n = \big((1-\alpha^{(N-n+1)/R})/(1-\alpha^{1/R})\big)^R$ and the optimal consumption is a fixed *fraction* of wealth at each date. Again the CRRA signature: everything scales with wealth.

§7 Brownian Motion

Discrete-time models are perfectly adequate for computation, but they are awkward: the answers depend on the step size, and the formulae are sums of binomial coefficients rather than anything one would want to differentiate. In this section we construct the continuous-time limit.

§7.1 From Random Walk to Brownian Motion

Return to the binomial model $S_n = S_0\xi_1\cdots\xi_n$ and take logarithms:

$$\log S_n = \log S_0 + \sum_{k=1}^n \log \xi_k,$$

a random walk. Now let each period represent a small time δ, so that $t = n\delta$, and choose the parameters so that the mean and variance accumulate at fixed rates:

$$\mathbb{E}[\log\xi] = \mu\delta, \qquad \mathrm{Var}(\log\xi) = \sigma^2\delta.$$

Then $\log S_t$ has mean $\log S_0 + \mu t$ and variance $\sigma^2 t$, independently of δ. Writing

$$W_t = \frac{\log(S_t/S_0) - \mu t}{\sigma},$$

the central limit theorem from Part IA suggests that, as $\delta \downarrow 0$ with t fixed, W_t should converge to a $N(0,t)$ random variable; and since the increments of the walk over disjoint blocks are independent, the same should hold for increments. The limiting object is Brownian motion.

Definition 7.1 (Brownian motion)

A **standard Brownian motion** is a stochastic process $(W_t)_{t\geq 0}$ such that

(i) $W_0 = 0$;

(ii) for $0 \leq s < t$, the increment $W_t - W_s$ is independent of $(W_u)_{0\leq u\leq s}$;

(iii) for $0 \leq s < t$, $W_t - W_s \sim N(0, t-s)$;

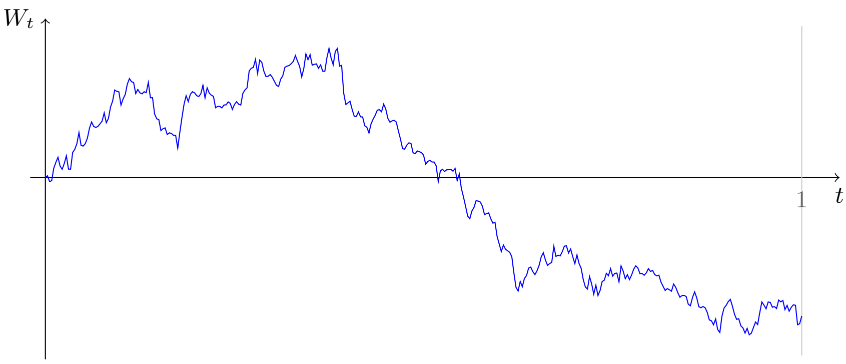
(iv) almost surely, $t \mapsto W_t$ is continuous.

That such a process exists is a genuine theorem, and not an easy one: one is asking for uncountably many random variables with prescribed joint laws whose sample paths are continuous.

Theorem 7.2 (Wiener, 1923)

Brownian motion exists.

We shall take this for granted. Here is a simulated path, drawn as an actual random walk with 360 steps on $[0, 1]$ – which, at this resolution, is visually indistinguishable from the real thing.



The path is continuous but visibly rough: it has no intervals of monotonicity, and in fact it is almost surely nowhere differentiable and of infinite variation on every interval. This roughness is not a defect of the model – it is precisely what makes the theory work, since it is the accumulated quadratic variation $\sum(W_{t_{i+1}} - W_{t_i})^2 \to t$ that produces the second-derivative term in Itô's formula.

§7.2 Basic Properties

Definition 7.3

A process $(X_t)_{t\geq 0}$ is **Gaussian** if $(X_{t_1}, \dots, X_{t_n})$ is a multivariate normal vector for all $0 \leq t_1 < \cdots < t_n$.

Theorem 7.4 (Gaussian characterisation)

A continuous process $(W_t)_{t\geq 0}$ with $W_0 = 0$ is a Brownian motion if and only if it is a Gaussian process with

$$\mathbb{E}[W_t] = 0 \qquad \text{and} \qquad \mathbb{E}[W_s W_t] = s \wedge t \quad \text{for all } s, t \geq 0.$$

Proof. ($\Rightarrow$) If W is a Brownian motion then $W_{t_1}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent normals, so any linear combination of $W_{t_1}, \dots, W_{t_n}$ is normal, hence W is Gaussian. Clearly $\mathbb{E}[W_t] = 0$, and for $s < t$,

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}[W_s(W_t - W_s)] = s + 0 = s.$$

($\Leftarrow$) Conversely, suppose W is Gaussian with the stated moments. For $s < t$, $W_t - W_s$ is normal (being a linear combination of jointly Gaussian variables) with mean 0 and variance

$$\mathbb{E}[W_t^2] - 2\mathbb{E}[W_s W_t] + \mathbb{E}[W_s^2] = t - 2s + s = t - s.$$

For $u < s < t$, $\mathbb{E}[(W_t - W_s)W_u] = u - u = 0$, so $W_t - W_s$ is uncorrelated with $(W_u)_{u \le s}$; as everything is jointly Gaussian, uncorrelated means independent (a fact from Part IA Probability). Hence W is a Brownian motion. □

This characterisation makes the following symmetries almost trivial to verify.

Theorem 7.5 (Invariance properties)
Let $(W_t)_{t \ge 0}$ be a Brownian motion. Then so are:

(i) (*Scaling*) $\tilde{W}_t = cW_{t/c^2}$, for any $c \neq 0$;

(ii) (*Simple Markov property*) $\tilde{W}_t = W_{t+T} - W_T$, for any fixed $T \ge 0$;

(iii) (*Time inversion*) $\tilde{W}_t = tW_{1/t}$ for $t > 0$, $\tilde{W}_0 = 0$;

(iv) (*Reflection*) $\tilde{W}_t = -W_t$.

Proof. Each is Gaussian with mean zero, so by Theorem 7.4 we need only check covariances. For (i), $\mathbb{E}[\tilde{W}_s \tilde{W}_t] = c^2 \,(s/c^2 \wedge t/c^2) = s \wedge t$. For (ii), $\mathbb{E}[\tilde{W}_s \tilde{W}_t] = (s{+}T) \wedge (t{+}T) - T = s \wedge t$. For (iii), taking $s < t$, $\mathbb{E}[\tilde{W}_s \tilde{W}_t] = st\,(1/s \wedge 1/t) = st/t = s$. For (iv) it is immediate. The only subtlety is continuity at 0 in (iii), which amounts to $W_t/t \to 0$ almost surely as $t \to \infty$; this is a form of the strong law of large numbers. □

Property (i) is worth dwelling on: Brownian motion is a fractal, statistically identical to itself under the rescaling $(t, w) \mapsto (c^2 t, cw)$. Space scales like the square root of time – which is the CLT again.

Theorem 7.6
Brownian motion is a Markov process: for bounded measurable g and $s < t$,

$$\mathbb{E}[g(W_t) \mid \mathcal{F}_s] = \mathbb{E}[g(W_t) \mid W_s],$$

where $\mathcal{F}_s = \sigma(W_u : u \le s)$.

Proof. Writing $W_t = (W_t - W_s) + W_s$ with $W_t - W_s \sim N(0, t - s)$ independent of $\mathcal{F}_s$,

$$\mathbb{E}[g(W_t) \mid \mathcal{F}_s] = \int_{\mathbb{R}} g(z + W_s) \frac{e^{-z^2/2(t-s)}}{\sqrt{2\pi(t-s)}} \,\mathrm{d}z,$$

which is a function of W_s alone. □

Later we shall want to restart the process not at a fixed time but at a random one, and for that the simple Markov property is not enough. The correct strengthening is the following, which we shall use without proof.

Theorem 7.7 (Strong Markov property)

Let W be a Brownian motion and T an almost surely finite stopping time for $(\mathcal{F}_t)$. Then

$$\left(W_{T+t} - W_T\right)_{t \geq 0}$$

is a Brownian motion, independent of $\mathcal{F}_T = \{A : A \cap \{T \leq t\} \in \mathcal{F}_t \text{ for all } t\}$.

Remark. The proof is a discretisation argument, and it is worth sketching because it shows exactly where continuity is used. Set $T_n = 2^{-n}\lceil 2^n T \rceil$, a stopping time taking countably many values $k2^{-n}$. On the event $\{T_n = k2^{-n}\} \in \mathcal{F}_{k2^{-n}}$ the simple Markov property applies, and summing over k shows that $(W_{T_n+t} - W_{T_n})_t$ is a Brownian motion independent of $\mathcal{F}_{T_n}$, hence of the smaller σ-algebra $\mathcal{F}_T$. Since $T_n \downarrow T$ and the paths are continuous, $W_{T_n+t} \to W_{T+t}$ almost surely, and the property passes to the limit. Without path continuity the last step fails, and indeed the strong Markov property is genuinely delicate for processes with jumps.

It is a good moment to make precise the roughness we noticed in the picture. The relevant quantity is the **quadratic variation**, and it is really the whole reason that stochastic calculus differs from ordinary calculus.

Theorem 7.8 (Quadratic variation)

Fix $t > 0$ and for each n let $0 = t_0^n < t_1^n < \cdots < t_{m_n}^n = t$ be a partition of $[0, t]$ with mesh $\delta_n = \max_i (t_i^n - t_{i-1}^n) \to 0$. Then

$$Q_n := \sum_{i=1}^{m_n} \left(W_{t_i^n} - W_{t_{i-1}^n}\right)^2 \longrightarrow t \qquad \text{in } L^2.$$

Proof. Write $\Delta_i = W_{t_i^n} - W_{t_{i-1}^n}$ and $h_i = t_i^n - t_{i-1}^n$, so that the Δ_i are independent with $\Delta_i \sim N(0, h_i)$. Then $\mathbb{E}[\Delta_i^2] = h_i$, so $\mathbb{E}[Q_n] = \sum_i h_i = t$ exactly. For the variance, independence gives

$$\operatorname{Var}(Q_n) = \sum_i \operatorname{Var}(\Delta_i^2) = \sum_i \left(\mathbb{E}[\Delta_i^4] - h_i^2\right) = \sum_i \left(3h_i^2 - h_i^2\right) = 2\sum_i h_i^2,$$

using $\mathbb{E}[Z^4] = 3$ for standard normal Z. Finally $\sum_i h_i^2 \leq \delta_n \sum_i h_i = \delta_n t \to 0$, so $\mathbb{E}[(Q_n - t)^2] \to 0$. □

The contrast with a differentiable function f is stark: there $\sum_i (f(t_i) - f(t_{i-1}))^2 \leq \delta_n \sum_i |f(t_i) - f(t_{i-1})| \to 0$, since the total variation is finite. So Brownian paths cannot be differentiable, and the following is immediate.

Corollary 7.9

Almost surely, the path $s \mapsto W_s$ has infinite total variation on $[0, t]$ for every $t > 0$; in particular it is not monotone on any interval, and is not of bounded variation, so $\int_0^t H_s \, \mathrm{d}W_s$ cannot be defined pathwise as a Stieltjes integral.

Proof. Along a subsequence, $Q_n \to t$ almost surely. If the path had total variation

$V < \infty$ on $[0, t]$ then, by uniform continuity of the path on the compact interval,

$$Q_n \leq \left(\max_i \left| W_{t_i^n} - W_{t_{i-1}^n} \right| \right) V \longrightarrow 0,$$

contradicting $Q_n \to t > 0$. □

The slogan is that $(\,\mathrm{d}W_t)^2 = \,\mathrm{d}t$: increments of Brownian motion are of size $\sqrt{\,\mathrm{d}t}$, so their squares accumulate at rate $\,\mathrm{d}t$ rather than vanishing. Everything unusual about Itô's formula comes from this one fact.

§7.3 Martingale Properties

Definition 7.10

A continuous process $(X_t)_{t\geq 0}$ adapted to a filtration $(\mathcal{F}_t)_{t\geq 0}$ is a **martingale** if each X_t is integrable and $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ for all $0 \leq s \leq t$.

Proposition 7.11

Let W be a Brownian motion and $(\mathcal{F}_t)$ its generated filtration. Then the following are martingales:

(i) W_t;

(ii) $W_t^2 - t$;

(iii) $\mathcal{E}_t^\theta := \exp(\theta W_t - \theta^2 t/2)$, for each fixed $\theta \in \mathbb{R}$.

Proof. In each case we use that $W_t - W_s$ is independent of $\mathcal{F}_s$ with law $N(0, t-s)$. (i) $\mathbb{E}[W_t \mid \mathcal{F}_s] = \mathbb{E}[W_t - W_s \mid \mathcal{F}_s] + W_s = 0 + W_s$. (ii) $\mathbb{E}[W_t^2 \mid \mathcal{F}_s] = \mathbb{E}[(W_t - W_s)^2 \mid \mathcal{F}_s] + 2W_s\mathbb{E}[W_t - W_s \mid \mathcal{F}_s] + W_s^2 = (t-s) + W_s^2$, and subtracting t gives the claim. (iii) $\mathbb{E}[e^{\theta(W_t - W_s)} \mid \mathcal{F}_s] = e^{\theta^2(t-s)/2}$ by the normal moment generating function, so

$$\mathbb{E}[\mathcal{E}_t^\theta \mid \mathcal{F}_s] = \mathcal{E}_s^\theta \, e^{-\theta^2(t-s)/2}\mathbb{E}[e^{\theta(W_t - W_s)} \mid \mathcal{F}_s] = \mathcal{E}_s^\theta.$$ □

The process $\mathcal{E}^\theta$ is the **exponential martingale**, and it is the single most useful object in this course: it is simultaneously the density of a change of measure, the discounted stock price under the risk-neutral measure, and a generating function for the moments of W.

Remarkably, each of (i)–(iii) characterises Brownian motion.

Theorem 7.12

A continuous process $(W_t)_{t\geq 0}$ with $W_0 = 0$ is a Brownian motion if and only if $(e^{\theta W_t - \theta^2 t/2})_{t \geq 0}$ is a martingale for every $\theta \in \mathbb{R}$.

Proof. Necessity is Proposition 7.11(iii). Conversely, the martingale property gives

$$\mathbb{E}\left[e^{\theta(W_t - W_s)} \mid \mathcal{F}_s\right] = e^{\theta^2(t-s)/2}$$

for every θ. The right-hand side is deterministic, and it is the moment generating function of $N(0, t-s)$; since moment generating functions determine distributions, $W_t - W_s \sim N(0, t-s)$ and, the conditional law being non-random, $W_t - W_s$ is independent of $\mathcal{F}_s$. $\square$

Theorem 7.13 (Lévy's characterisation)

A continuous process $(W_t)_{t \geq 0}$ with $W_0 = 0$ is a Brownian motion if and only if both (W_t) and $(W_t^2 - t)$ are martingales.

We omit the proof, which is a genuine piece of stochastic analysis. It is a strikingly strong statement: two moment conditions plus continuity pin the process down completely.

§7.4 Hitting Times and the Reflection Principle

Definition 7.14

For $a \in \mathbb{R}$, let $T_a = \inf\{t \geq 0 : W_t = a\}$ be the first **hitting time** of level a, and let $M_t = \max_{0 \leq s \leq t} W_s$ be the running maximum.

Note that $\{T_a \leq t\} = \{M_t \geq a\}$ for $a > 0$, by continuity of paths.

Theorem 7.15

For every $a \in \mathbb{R} \setminus \{0\}$, $T_a < \infty$ almost surely.

Proof. By symmetry take $a > 0$; we show $Z := \sup_{t \geq 0} W_t = \infty$ a.s. By the scaling property, for any $c > 0$ the process $\tilde{W}_t = cW_{t/c^2}$ is a Brownian motion, so

$$\mathbb{P}\Big(\sup_{t \geq 0} W_t > a\Big) = \mathbb{P}\Big(\sup_{t \geq 0} \tilde{W}_t > a\Big) = \mathbb{P}\Big(\sup_{t \geq 0} W_t > a/c\Big).$$

Letting $c \downarrow 0$ shows that $\mathbb{P}(Z > a)$ is the same for all $a > 0$, so $Z \in \{0, \infty\}$ almost surely. Now let $\hat{Z} = \sup_{t \geq 1}(W_t - W_1)$, which has the same law as Z and is independent of W_1. If $p = \mathbb{P}(Z = 0)$ then, since $\{Z = 0\} \subseteq \{W_1 \leq 0\} \cap \{\hat{Z} = 0\}$,

$$p = \mathbb{P}(Z = 0) \leq \mathbb{P}(W_1 \leq 0)\, \mathbb{P}(\hat{Z} = 0) = \tfrac{1}{2} p,$$

forcing $p = 0$. Hence $Z = \infty$ a.s. $\square$

The key tool for computing with M_t is the following, which says that if we reflect the path in the level a after it first reaches a, we get another Brownian motion.

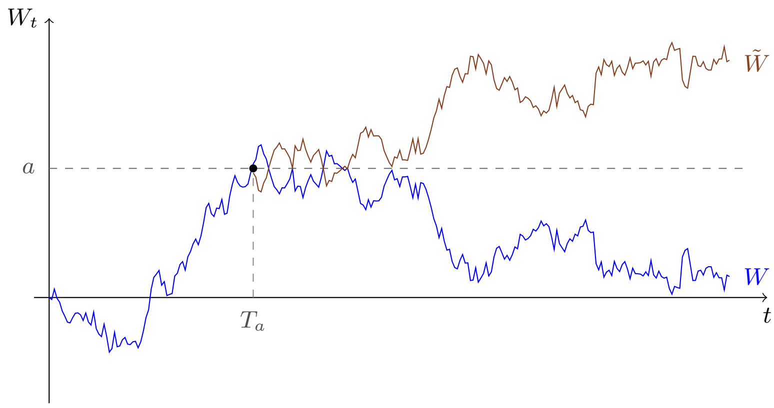
Theorem 7.16 (Reflection principle)

Let W be a Brownian motion and $a > 0$. Then

$$\tilde{W}_t = W_t \mathbb{1}_{\{t \leq T_a\}} + (2a - W_t) \mathbb{1}_{\{t > T_a\}}$$

is a Brownian motion.

Proof. By Theorem 7.15 the time T_a is an almost surely finite stopping time, so the strong Markov property (Theorem 7.7) applies: $(W_{T_a+t} - W_{T_a})_{t\geq 0} = (W_{T_a+t} - a)_{t\geq 0}$ is a Brownian motion independent of $\mathcal{F}_{T_a}$. By the reflection invariance, so is its negative. Since $\tilde{W}_{T_a+t} = a - (W_{T_a+t} - a)$, the process $\tilde{W}$ is obtained from W by replacing the post-T_a increments by their negatives, which does not change the law. □



Lemma 7.17

For bounded measurable g and $a > 0$,

$$\mathbb{E}\big[g(W_t)\mathbb{1}_{\{W_t\leq a,\ M_t\geq a\}}\big] = \mathbb{E}\big[g(2a - W_t)\mathbb{1}_{\{W_t\geq a\}}\big].$$

Proof. Let $\tilde{W}$ be as in Theorem 7.16 and $\tilde{M}_t = \max_{s\leq t} \tilde{W}_s$. Since reflection does not change whether the level a is reached, $\{M_t \geq a\} = \{\tilde{M}_t \geq a\}$; and on that event $W_t = 2a - \tilde{W}_t$. Hence

$$\begin{aligned}\mathbb{E}\big[g(W_t)\mathbb{1}_{\{W_t\leq a, M_t\geq a\}}\big] &= \mathbb{E}\big[g(2a - \tilde{W}_t)\mathbb{1}_{\{\tilde{W}_t\geq a,\ \tilde{M}_t\geq a\}}\big] \\ &= \mathbb{E}\big[g(2a - \tilde{W}_t)\mathbb{1}_{\{\tilde{W}_t\geq a\}}\big],\end{aligned}$$

because $\{\tilde{W}_t \geq a\} \subseteq \{\tilde{M}_t \geq a\}$. As $\tilde{W}$ has the same law as W, we may drop the tildes. □

Corollary 7.18

For $a \geq b$ and $a \geq 0$, $\mathbb{P}(W_t \leq b,\ M_t \geq a) = \mathbb{P}(W_t \geq 2a - b)$.

Proof. Take $g(w) = \mathbb{1}_{\{w\leq b\}}$ in Lemma 7.17: the left side is $\mathbb{P}(W_t \leq b, M_t \geq a)$ (using $b \leq a$), and the right side is $\mathbb{P}(2a - W_t \leq b,\ W_t \geq a) = \mathbb{P}(W_t \geq 2a - b)$, the constraint $W_t \geq a$ being implied. □

This immediately gives the joint law of (M_t, W_t) and the law of the hitting time.

Theorem 7.19

For $t > 0$:

(i) M_t has the same law as $|W_t|$; explicitly $\mathbb{P}(M_t \geq a) = 2\Phi(-a/\sqrt{t})$ for $a \geq 0$;

(ii) the pair (M_t, W_t) has joint density, for $a \geq b^+$,

$$f_{M_t,W_t}(a,b) = \frac{2(2a-b)}{t^{3/2}\sqrt{2\pi}} \exp\left(-\frac{(2a-b)^2}{2t}\right);$$

(iii) the hitting time T_a ($a > 0$) has density

$$f_{T_a}(t) = \frac{a}{t^{3/2}\sqrt{2\pi}} e^{-a^2/2t}, \qquad t > 0,$$

and $\mathbb{E}[T_a] = \infty$.

Proof. (i) Splitting according to the value of W_t and using Corollary 7.18 with $b = a$,

$$\begin{aligned}\mathbb{P}(M_t \geq a) &= \mathbb{P}(M_t \geq a,\, W_t \geq a) + \mathbb{P}(M_t \geq a,\, W_t \leq a)\\ &= \mathbb{P}(W_t \geq a) + \mathbb{P}(W_t \geq a) = 2\Phi(-a/\sqrt{t}),\end{aligned}$$

where the first term used $\{W_t \geq a\} \subseteq \{M_t \geq a\}$. This is exactly $\mathbb{P}(|W_t| \geq a)$.

(ii) Differentiate $\mathbb{P}(W_t \leq b, M_t \geq a) = \mathbb{P}(W_t \geq 2a - b) = 1 - \Phi((2a-b)/\sqrt{t})$ once in a and once in b; the stated formula results.

(iii) By continuity of paths $\{T_a \leq t\} = \{M_t \geq a\}$, so $\mathbb{P}(T_a \leq t) = 2\Phi(-a/\sqrt{t})$; differentiating in t gives the density. Since $f_{T_a}(t) \sim a/(\sqrt{2\pi}t^{3/2})$ as $t \to \infty$, we get $\int^\infty t f_{T_a}(t)\,\mathrm{d}t = \infty$. □

So Brownian motion hits every level with probability one, but takes infinitely long on average to do so – the same phenomenon we saw for the recurrent random walk in Example 5.20, and for exactly the same reason.

§7.5 The Cameron–Martin Theorem

Finally we need to be able to change measure so as to add a drift to Brownian motion. The finite-dimensional version is the Gaussian tilt of Example 4.4: if $Z \sim N(0,1)$ and $\alpha \in \mathbb{R}$ then for bounded f,

$$\mathbb{E}\big[e^{\alpha Z - \alpha^2/2} f(Z)\big] = \mathbb{E}[f(Z+\alpha)],$$

and in n dimensions, for $Z \sim N_n(0, I)$ and $\alpha \in \mathbb{R}^n$,

$$\mathbb{E}\big[e^{\alpha^\top Z - \|\alpha\|^2/2} f(Z)\big] = \mathbb{E}[f(Z+\alpha)].$$

Theorem 7.20 (Cameron–Martin)
Let W be a Brownian motion, $c \in \mathbb{R}$ and $t > 0$. Then for suitably integrable functionals g on $C([0,t])$,

$$\mathbb{E}\big[g\big((W_s + cs)_{0\leq s\leq t}\big)\big] = \mathbb{E}\big[e^{cW_t - c^2t/2} g\big((W_s)_{0\leq s\leq t}\big)\big].$$

Proof. By a monotone class argument it suffices to treat $g(w) = G(w(t_1), \dots, w(t_n))$ for $0 < t_1 < \cdots < t_n = t$. Writing $Z_k = (W_{t_k} - W_{t_{k-1}})/\sqrt{t_k - t_{k-1}}$ (with $t_0 = 0$), the Z_k are i.i.d. $N(0,1)$, and g may be rewritten as $H(Z_1, \dots, Z_n)$ for a suitable H. Adding the drift cs shifts Z_k by $\alpha_k = c\sqrt{t_k - t_{k-1}}$, so

$$\begin{aligned}\mathbb{E}\big[g((W_s + cs)_{s\le t})\big] &= \mathbb{E}\big[H(Z_1 + \alpha_1, \dots, Z_n + \alpha_n)\big] \\ &= \mathbb{E}\Big[\exp\Big(\sum_k \alpha_k Z_k - \tfrac{1}{2}\sum_k \alpha_k^2\Big) H(Z_1, \dots, Z_n)\Big] \\ &= \mathbb{E}\big[e^{cW_t - c^2 t/2}\, g((W_s)_{s\le t})\big],\end{aligned}$$

since $\sum_k \alpha_k Z_k = c\sum_k (W_{t_k} - W_{t_{k-1}}) = cW_t$ and $\sum_k \alpha_k^2 = c^2 t$. □

Corollary 7.21 (Girsanov, for constant drift)

Fix $c \in \mathbb{R}$ and a horizon $T > 0$, and define $\mathbb{Q} \sim \mathbb{P}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = e^{cW_T - c^2 T/2}.$$

Then $\tilde{W}_t = W_t - ct$, $0 \le t \le T$, is a $\mathbb{Q}$-Brownian motion.

Proof. For a bounded functional g,

$$\mathbb{E}_{\mathbb{Q}}\big[g((\tilde{W}_t)_{t\le T})\big] = \mathbb{E}_{\mathbb{P}}\big[e^{cW_T - c^2T/2} g((W_t - ct)_{t\le T})\big] = \mathbb{E}_{\mathbb{P}}\big[g((W_t)_{t\le T})\big]$$

by Theorem 7.20 applied with drift $-c$. So the law of $\tilde{W}$ under $\mathbb{Q}$ equals the law of W under $\mathbb{P}$. □

Notice what has happened: changing the measure changes the *drift* of the process, but not its volatility. This is the continuous-time avatar of the fact that the risk-neutral measure changes the expected return of a stock but not the size of its fluctuations, and it is the reason why the Black–Scholes price does not depend on μ.

§8 The Black–Scholes Model

We are finally in a position to write down the model which made mathematical finance a subject: a single stock whose price is a geometric Brownian motion, and a bank account paying a constant continuously compounded rate.

§8.1 The Model

Recall the passage to the limit in §7.1, where we found $\log S_t = \log S_0 + \mu t + \sigma W_t$ in the small-step limit. We take this as our definition.

Definition 8.1 (Black–Scholes model)

Fix constants $\mu \in \mathbb{R}$ (the **drift**), $\sigma > 0$ (the **volatility**) and $r \in \mathbb{R}$ (the interest rate), and let W be a Brownian motion. The **Black–Scholes model** has a bank account worth e^{rt} at time t and a stock whose price is the **geometric Brownian**

motion

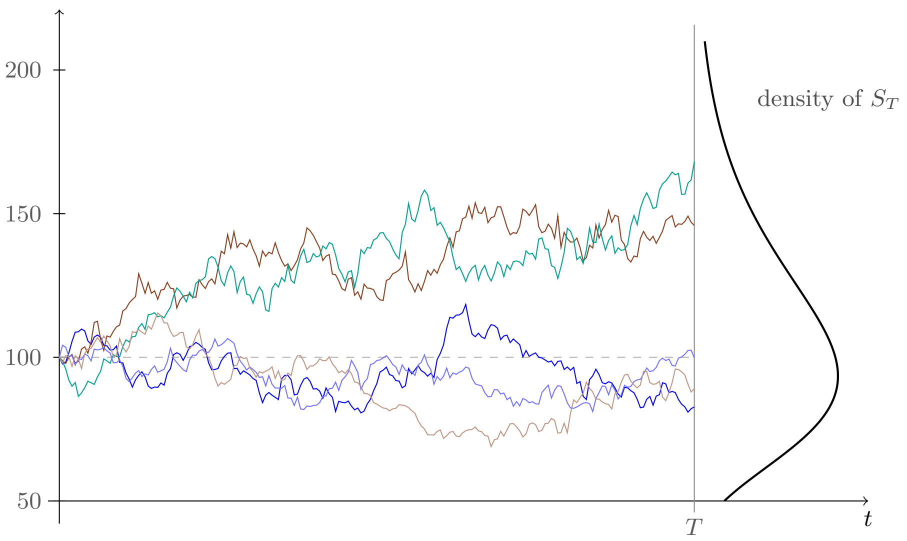
$$S_t = S_0\, e^{\mu t + \sigma W_t}.$$

The interest rate convention is the limit of the discrete one: $(1+r\delta)^{t/\delta} \to e^{rt}$ as $\delta \downarrow 0$, so a continuously compounded rate r turns one pound into e^{rt} pounds.

Some immediate consequences. Since $W_t \sim N(0,t)$, the price S_t is **lognormal**: $\log S_t \sim N(\log S_0 + \mu t, \sigma^2 t)$. In particular $S_t > 0$ always, which is as it should be for a limited-liability share, and

$$\mathbb{E}[S_t] = S_0 e^{(\mu+\sigma^2/2)t}.$$

Note the $\sigma^2/2$: the expected price grows faster than the typical price, because the lognormal distribution is skewed. Here are five simulated paths with $S_0 = 100$, $\mu = 0.06$, $\sigma = 0.25$ over two years, together with the density of S_T at the terminal time.



The right-hand curve is the lognormal density of S_T, drawn sideways so that heights correspond to prices. Note its asymmetry: a long right tail, and a floor at zero.

§8.2 The Risk-Neutral Measure

Proposition 8.2

The Black–Scholes model on $[0,T]$ has a risk-neutral measure: there is $\mathbb{Q} \sim \mathbb{P}$ under which $(e^{-rt}S_t)_{0\le t\le T}$ is a martingale. Explicitly, with

$$c = \frac{r-\mu}{\sigma} - \frac{\sigma}{2}, \qquad \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = e^{cW_T - c^2T/2},$$

the process $\tilde{W}_t = W_t - ct$ is a $\mathbb{Q}$-Brownian motion and

$$S_t = S_0 \exp\left(\left(r - \tfrac{\sigma^2}{2}\right)t + \sigma\tilde{W}_t\right).$$

Proof. By Corollary 7.21, $\tilde{W}_t = W_t - ct$ is a $\mathbb{Q}$-Brownian motion. Substituting $W_t = \tilde{W}_t + ct$ into the definition of S_t,

$$S_t = S_0 e^{\mu t + \sigma(\tilde{W}_t + ct)} = S_0 e^{(\mu+\sigma c)t + \sigma\tilde{W}_t} = S_0 e^{(r-\sigma^2/2)t + \sigma\tilde{W}_t},$$

since $\mu + \sigma c = r - \sigma^2/2$. Hence $e^{-rt}S_t = S_0 e^{\sigma\tilde{W}_t - \sigma^2 t/2}$, which is the exponential martingale of Proposition 7.11 for the $\mathbb{Q}$-Brownian motion $\tilde{W}$, with $\theta = \sigma$. $\square$

Two things deserve comment. First, the drift μ has vanished from the risk-neutral dynamics: under $\mathbb{Q}$ the stock grows at the riskless rate r, whatever it does in reality. Second, the volatility σ is unchanged – changing measure can shift a Brownian motion by a drift but cannot alter its scale. So of the two parameters of the model, only σ will appear in any option price.

Definition 8.3 (Black–Scholes price)

Let Y be the payout of a European claim with maturity T. Its **Black–Scholes price** at time t is

$$\pi_t = e^{-r(T-t)}\,\mathbb{E}_{\mathbb{Q}}\big[Y \mid \mathcal{F}_t\big].$$

By construction $(e^{-rt}\pi_t)$ is a $\mathbb{Q}$-martingale, so the augmented market has no arbitrage. In fact the Black–Scholes model is complete, so this is the *only* arbitrage-free price; we prove this in §8.5, once we have Itô's lemma to hand.

§8.3 The Black–Scholes Formula

Let $Y = g(S_T)$ be vanilla and write $\tau = T - t$ for the time to maturity. Under $\mathbb{Q}$,

$$S_T = S_t \exp\Big(\big(r - \tfrac{\sigma^2}{2}\big)\tau + \sigma(\tilde{W}_T - \tilde{W}_t)\Big),$$

with $\tilde{W}_T - \tilde{W}_t \sim N(0,\tau)$ independent of $\mathcal{F}_t$. So $\pi_t = V(t, S_t)$ where

$$\boxed{V(t,s) = e^{-r\tau}\,\mathbb{E}\Big[g\Big(s\,e^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}Z}\Big)\Big] = e^{-r\tau}\int_{-\infty}^{\infty} g\Big(se^{(r-\sigma^2/2)\tau+\sigma\sqrt{\tau}z}\Big)\frac{e^{-z^2/2}}{\sqrt{2\pi}}\,\mathrm{d}z}$$

for $Z \sim N(0,1)$. This is a completely explicit one-dimensional integral. For the call it evaluates in closed form.

Theorem 8.4 (Black–Scholes formula)

The price of a European call with strike K and maturity T is $V(t, S_t)$ where, with $\tau = T - t$,

$$V(t,s) = s\,\Phi(d_1) - Ke^{-r\tau}\Phi(d_2),$$

$$d_{1,2} = \frac{\log(s/K) + \big(r \pm \frac{\sigma^2}{2}\big)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau},$$

and Φ is the standard normal distribution function.

Proof. Write $g(s) = (s-K)^+$ and $S_T = se^{(r-\sigma^2/2)\tau+\sigma\sqrt{\tau}Z}$. The option finishes in the money precisely when

$$se^{(r-\sigma^2/2)\tau+\sigma\sqrt{\tau}z} > K \iff z > \frac{\log(K/s) - (r-\sigma^2/2)\tau}{\sigma\sqrt{\tau}} = -d_2.$$

Hence

$$V(t,s) = e^{-r\tau} \int_{-d_2}^{\infty} \left(se^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}z} - K \right) \varphi(z)\, \mathrm{d}z,$$

where φ is the standard normal density. The second piece is $-e^{-r\tau}K\big(1-\Phi(-d_2)\big) = -Ke^{-r\tau}\Phi(d_2)$. For the first, note the completing-the-square identity

$$e^{-\sigma^2\tau/2+\sigma\sqrt{\tau}z}\varphi(z) = \varphi\big(z-\sigma\sqrt{\tau}\big),$$

so that

$$\begin{aligned} e^{-r\tau} \int_{-d_2}^{\infty} se^{(r-\sigma^2/2)\tau+\sigma\sqrt{\tau}z}\varphi(z)\, \mathrm{d}z &= s\int_{-d_2}^{\infty} \varphi(z-\sigma\sqrt{\tau})\, \mathrm{d}z \\ &= s\int_{-d_2-\sigma\sqrt{\tau}}^{\infty} \varphi(u)\, \mathrm{d}u = s\,\Phi(d_2+\sigma\sqrt{\tau}), \end{aligned}$$

and $d_2 + \sigma\sqrt{\tau} = d_1$. □

Corollary 8.5 (Put–call parity)

The price of the corresponding European put is

$$P(t,s) = Ke^{-r\tau}\Phi(-d_2) - s\Phi(-d_1), \qquad C(t,s) - P(t,s) = s - Ke^{-r\tau}.$$

Proof. As in Proposition 6.13, $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, so

$$C - P = e^{-r\tau}\mathbb{E}_{\mathbb{Q}}[S_T - K \mid \mathcal{F}_t] = s - Ke^{-r\tau}$$

because $e^{-rt}S_t$ is a $\mathbb{Q}$-martingale. Substituting the formula for C and using $\Phi(x) = 1 - \Phi(-x)$ gives the expression for P. □

Example 8.6 (A numerical example)

Take $S_t = 100$, $K = 100$, $r = 0.05$, $\sigma = 0.2$ and $\tau = 1$ year. Then

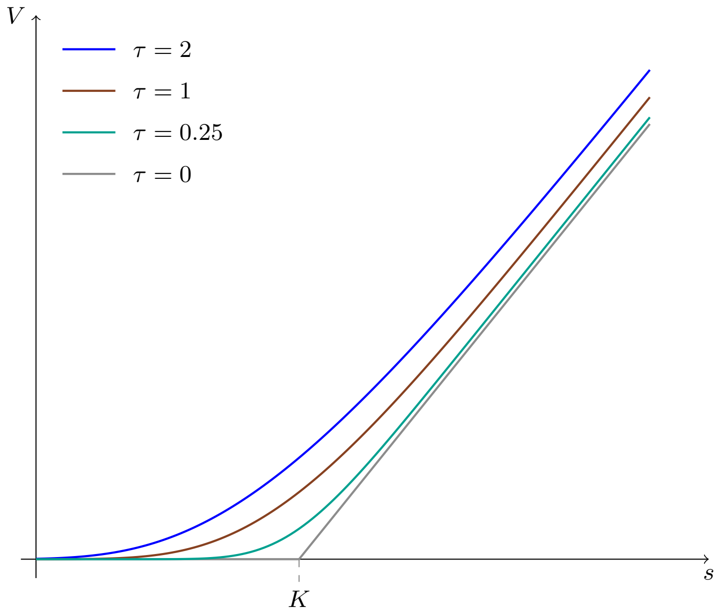
$$d_1 = \frac{0 + (0.05 + 0.02)\cdot 1}{0.2} = 0.35, \qquad d_2 = 0.35 - 0.2 = 0.15,$$

and $\Phi(0.35) = 0.6368$, $\Phi(0.15) = 0.5596$, $e^{-0.05} = 0.9512$. So

$$C = 100(0.6368) - 100(0.9512)(0.5596) = 63.68 - 53.23 = 10.45.$$

The corresponding put is $P = C - S + Ke^{-r\tau} = 10.45 - 100 + 95.12 = 5.57$. The hedge ratio is $\Delta = \Phi(d_1) = 0.637$: on selling this call one immediately buys 0.637 shares.

The general shape of V as a function of s is worth internalising. As $\tau \downarrow 0$ the price converges to the payout $(s-K)^+$; for larger τ the curve is a smoothed version of that hockey stick, lying above it, and increasing in τ.



§8.4 Itô's Lemma and the Black–Scholes PDE

The derivation above is the modern one. Black and Scholes argued differently, via a partial differential equation, and that route is both historically important and computationally useful. It rests on the fundamental rule of stochastic calculus.

Theorem 8.7 (Itô's lemma)
Let X satisfy the stochastic differential equation

$$\mathrm{d}X_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}W_t,$$

and let $f(t,x)$ be twice continuously differentiable. Then

$$\mathrm{d}f(t,X_t) = \left(\frac{\partial f}{\partial t} + a_t \frac{\partial f}{\partial x} + \frac{1}{2} b_t^2 \frac{\partial^2 f}{\partial x^2}\right)\,\mathrm{d}t + b_t \frac{\partial f}{\partial x}\,\mathrm{d}W_t,$$

all partial derivatives being evaluated at (t, X_t).

We take this as given – it is proved in Part III. The heuristic is that $(\mathrm{d}W_t)^2 = \mathrm{d}t$: Taylor expanding f to second order, the term $\frac{1}{2} f_{xx}(\mathrm{d}X_t)^2 = \frac{1}{2} f_{xx} b_t^2 (\mathrm{d}W_t)^2$ is of order $\mathrm{d}t$ and must be retained, because by Theorem 7.8 the quadratic variation of Brownian motion over $[0,t]$ is t rather than 0. This is precisely the roughness of the Brownian path made useful.

Proposition 8.8
The geometric Brownian motion $S_t = S_0 e^{\mu t + \sigma W_t}$ satisfies

$$\mathrm{d}S_t = \left(\mu + \tfrac{\sigma^2}{2}\right) S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t.$$

Under $\mathbb{Q}$ this reads $\mathrm{d}S_t = rS_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}\tilde{W}_t$.

Proof. Apply Itô with $X_t = W_t$ (so $a = 0$, $b = 1$) and $f(t, x) = S_0 e^{\mu t + \sigma x}$: the partial derivatives are $f_t = \mu f$, $f_x = \sigma f$, $f_{xx} = \sigma^2 f$, giving $\mathrm{d}S_t = (\mu + \sigma^2/2) S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t$. For the second statement substitute $W_t = \tilde{W}_t + ct$ and use $\mu + \sigma c + \sigma^2/2 = r$. □

So the parameter $\mu + \sigma^2/2$ is the instantaneous expected rate of return, and σ is the instantaneous standard deviation per unit time.

Theorem 8.9 (Black–Scholes PDE)
Let $V(t, s)$ be the Black–Scholes price of a vanilla claim with payout $g(S_T)$. Then V solves

$$\frac{\partial V}{\partial t} + rs\frac{\partial V}{\partial s} + \frac{1}{2}\sigma^2 s^2 \frac{\partial^2 V}{\partial s^2} = rV, \qquad V(T, s) = g(s).$$

Proof. Suppose a self-financing portfolio holds Δ_t shares and the rest in the bank, with value X_t. Then

$$\mathrm{d}X_t = r(X_t - \Delta_t S_t)\,\mathrm{d}t + \Delta_t \,\mathrm{d}S_t.$$

On the other hand, applying Itô to $V(t, S_t)$ with $\mathrm{d}S_t = (\mu + \sigma^2/2) S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t$,

$$\mathrm{d}V(t, S_t) = \Big(V_t + \big(\mu + \tfrac{\sigma^2}{2}\big) S_t V_s + \tfrac{1}{2}\sigma^2 S_t^2 V_{ss}\Big)\,\mathrm{d}t + \sigma S_t V_s \,\mathrm{d}W_t.$$

If the portfolio is to replicate the claim, we need $X_t = V(t, S_t)$ for all t, so the two expressions must agree. Matching the $\mathrm{d}W_t$ terms forces

$$\Delta_t = V_s(t, S_t),$$

and then matching the $\mathrm{d}t$ terms, after substituting $\mathrm{d}S_t$ into the first expression, gives

$$rV - rS_t V_s = V_t + \tfrac{1}{2}\sigma^2 S_t^2 V_{ss},$$

which rearranges to the stated PDE. The terminal condition is $V(T, s) = g(s)$, since at maturity the portfolio must be worth the payout. □

Notice again the disappearance of μ: the drift terms cancelled when we chose $\Delta_t = V_s$. This is the mathematical content of the phrase 'delta hedging removes the risk'.

Remark (The binomial heuristic)**.** One can also derive the PDE directly as a limit of the binomial recursion, which is illuminating because it shows where each term comes from. Take a time step δ and set

$$1 + b = e^{\mu\delta + \sigma\sqrt{\delta}}, \quad 1 + a = e^{\mu\delta - \sigma\sqrt{\delta}}, \quad r_{\text{bin}} = r\delta,$$

so that $b = \sigma\sqrt{\delta} + (\mu + \frac{\sigma^2}{2})\delta + O(\delta^{3/2})$ and similarly for a. The risk-neutral probability is $q = (r\delta - a)/(b - a)$, and one computes

$$qb + (1 - q)a = r\delta, \qquad qb^2 + (1 - q)a^2 = \sigma^2\delta + O(\delta^{3/2}).$$

Now expand the binomial recursion $(1 + r\delta)V(t - \delta, s) = qV(t, s(1 + b)) + (1 - q)V(t, s(1 + a))$ to second order in δ:

$$V + \Big(rV - \frac{\partial V}{\partial t}\Big)\delta = V + \frac{\partial V}{\partial s} s\big(qb + (1 - q)a\big) + \frac{1}{2}\frac{\partial^2 V}{\partial s^2} s^2\big(qb^2 + (1 - q)a^2\big) + o(\delta)$$

$$= V + \frac{\partial V}{\partial s} sr\delta + \frac{1}{2}\frac{\partial^2 V}{\partial s^2} s^2\sigma^2\delta + o(\delta).$$

Cancelling V, dividing by δ and letting $\delta \to 0$ gives the Black–Scholes PDE. Note that the first-order term picks out the interest rate and the second-order term the volatility – exactly as in Itô's lemma.

Remark (Reduction to the heat equation)**.** The Black–Scholes PDE is the heat equation in disguise. If $u(\tau, x) = \mathbb{E}[f(x + \sqrt{\tau}Z)]$ for $Z \sim N(0,1)$ then differentiating under the expectation gives $u_\tau = \frac{1}{2}u_{xx}$. Setting

$$u(\tau, x) = e^{r\tau}\, V\Big(T - \frac{\tau}{\sigma^2},\ e^{x-(r/\sigma^2-1/2)\tau}\Big),$$

one checks that V solves the Black–Scholes PDE if and only if u solves $u_\tau = \frac{1}{2}u_{xx}$ with $u(0,x) = g(e^x)$. This is why Black–Scholes prices behave like diffusing heat: the payoff's kink is smoothed out instantly and increasingly with time to maturity.

§8.5 Replication and Completeness

We can now discharge the promise made after Proposition 8.2 and show that the Black–Scholes price is not merely *an* arbitrage-free price but the *only* one, because the claim can be replicated exactly.

The continuous-time analogue of the self-financing condition of §6.1 is obtained by letting the rebalancing interval shrink. If the agent holds θ_t shares and θ_t^0 pounds in the bank, with wealth $X_t = \theta_t^0 + \theta_t S_t$, then over $[t, t + \mathrm{d}t]$ no money enters or leaves, so the change in wealth is entirely due to the change in the prices of the assets held:

Definition 8.10 (Self-financing, continuous time)
An adapted pair (θ_t^0, θ_t) is **self-financing** if the wealth $X_t = \theta_t^0 + \theta_t S_t$ satisfies

$$\mathrm{d}X_t = r\theta_t^0\, \mathrm{d}t + \theta_t\, \mathrm{d}S_t = r\big(X_t - \theta_t S_t\big)\, \mathrm{d}t + \theta_t\, \mathrm{d}S_t.$$

Equivalently, writing $\tilde{X}_t = e^{-rt}X_t$ and $\tilde{S}_t = e^{-rt}S_t$, the discounted wealth satisfies $\mathrm{d}\tilde{X}_t = \theta_t\, \mathrm{d}\tilde{S}_t$.

The equivalence is a one-line application of Itô to $e^{-rt}X_t$, and it is the exact analogue of the discrete identity $\tilde{X}_n = X_0 + (\theta \cdot \tilde{S})_n$: in the numeraire of the bank account, a self-financing wealth process is a stochastic integral against the discounted prices, with no drift of its own.

Theorem 8.11 (Completeness of the Black–Scholes model)
Let g be such that $V(t,s)$, defined by the risk-neutral expectation of §8.3, is $C^{1,2}$ on $[0,T) \times (0,\infty)$ with bounded V_s. Then the self-financing strategy with initial wealth $X_0 = V(0, S_0)$ and

$$\theta_t = V_s(t, S_t), \qquad \theta_t^0 = V(t, S_t) - S_t V_s(t, S_t),$$

satisfies $X_t = V(t, S_t)$ for every $t \le T$; in particular $X_T = g(S_T)$ almost surely.

Hence every such claim is replicable and the Black–Scholes price is the unique arbitrage-free price.

Proof. By Theorem 8.9, V solves the Black–Scholes PDE. Apply Itô's lemma to $V(t, S_t)$ under $\mathbb{Q}$, where $\mathrm{d}S_t = rS_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}\tilde{W}_t$:

$$\mathrm{d}V(t, S_t) = \left(V_t + rS_tV_s + \tfrac{1}{2}\sigma^2 S_t^2 V_{ss}\right) \mathrm{d}t + \sigma S_t V_s \,\mathrm{d}\tilde{W}_t = rV(t, S_t)\,\mathrm{d}t + \sigma S_t V_s \,\mathrm{d}\tilde{W}_t,$$

the drift having been replaced using the PDE. On the other hand the self-financing wealth process with $\theta_t = V_s(t, S_t)$ satisfies

$$\mathrm{d}X_t = r\big(X_t - \theta_t S_t\big)\,\mathrm{d}t + \theta_t\big(rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}\tilde{W}_t\big) = rX_t\,\mathrm{d}t + \sigma S_t V_s\,\mathrm{d}\tilde{W}_t.$$

So $D_t = X_t - V(t, S_t)$ satisfies $\mathrm{d}D_t = rD_t\,\mathrm{d}t$ with $D_0 = 0$, whence $D_t = D_0e^{rt} = 0$ for all t. Continuity of V up to $t = T$ gives $X_T = V(T, S_T) = g(S_T)$.

For uniqueness: if π were another arbitrage-free price for the claim, then selling at the higher of π and $V(0, S_0)$, buying at the lower, and running the replicating strategy would produce a riskless profit. □

Remark. The hypotheses are ugly and can be removed. The clean statement is the **martingale representation theorem**: if $(\mathcal{F}_t)$ is the filtration generated by a Brownian motion $\tilde{W}$ and M is a square-integrable $\mathbb{Q}$-martingale, then there is a previsible H with $M_t = M_0 + \int_0^t H_s\,\mathrm{d}\tilde{W}_s$. Applying this to $M_t = \mathbb{E}_{\mathbb{Q}}[e^{-rT}Y \mid \mathcal{F}_t]$ and setting $\theta_t = H_t/(\sigma\tilde{S}_t)$ replicates *any* integrable $\mathcal{F}_T$-measurable claim, with no smoothness assumed. This is the exact continuous-time analogue of the second fundamental theorem: the risk-neutral measure is unique, so the model is complete.

§8.6 The Greeks and Hedging

The partial derivatives of V have names and are collectively called the **Greeks**. They measure the sensitivity of the price to each input, and a trading desk manages a book by keeping them within limits.

Definition 8.12 (The Greeks)
For a claim with price $V(t, s)$, define

$$\Delta = \frac{\partial V}{\partial s}, \quad \Gamma = \frac{\partial^2 V}{\partial s^2}, \quad \Theta = \frac{\partial V}{\partial t}, \quad \mathcal{V} = \frac{\partial V}{\partial \sigma}, \quad \rho = \frac{\partial V}{\partial r},$$

called **delta**, **gamma**, **theta**, **vega** and **rho**. Holding Δ units of the stock is called **delta hedging**.

(The name 'vega' is not a Greek letter; the convention is that when one runs out of Greek letters one uses the names of stars, or television celebrities.)

Proposition 8.13

For a European call in the Black–Scholes model,

$$\Delta = \Phi(d_1), \qquad \Gamma = \frac{\varphi(d_1)}{s\sigma\sqrt{\tau}}, \qquad \mathcal{V} = s\sqrt{\tau}\,\varphi(d_1), \qquad \rho = K\tau e^{-r\tau}\Phi(d_2),$$

$$\Theta = -\frac{s\sigma\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2).$$

Proof. Differentiating $V = s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$ in s and using $\partial d_1/\partial s = \partial d_2/\partial s = 1/(s\sigma\sqrt{\tau})$,

$$\Delta = \Phi(d_1) + \frac{s\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)}{s\sigma\sqrt{\tau}}.$$

The bracketed term vanishes, by the identity $s\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, which follows from $d_2 = d_1 - \sigma\sqrt{\tau}$:

$$\frac{\varphi(d_1)}{\varphi(d_2)} = e^{-(d_1^2-d_2^2)/2} = e^{-\sigma\sqrt{\tau}\,(d_1-\sigma\sqrt{\tau}/2)} = e^{-\log(s/K)-r\tau} = \frac{Ke^{-r\tau}}{s}.$$

Hence $\Delta = \Phi(d_1)$, and differentiating once more gives Γ. The remaining Greeks follow similarly, using the same identity to cancel the terms coming from differentiating d_1 and d_2; for Θ one uses in addition $\partial d_1/\partial t = -\partial d_1/\partial\tau$ and $\partial(d_1-d_2)/\partial\tau = \sigma/(2\sqrt{\tau})$. $\square$

Two sanity checks on these formulae are worth making. First, all of $\Delta, \Gamma, \mathcal{V}, \rho$ are positive and Θ is negative: a call is worth more if the stock rises, if volatility rises, if rates rise, and less as time runs out. Second, the Greeks are not independent, because the Black–Scholes PDE of Theorem 8.9 is precisely the relation

$$\Theta + rs\Delta + \tfrac{1}{2}\sigma^2 s^2\Gamma = rV$$

among them. In practice this is how a desk thinks about the trade-off it faces: a hedged book is delta-neutral, so $\Theta = rV - \frac{1}{2}\sigma^2 s^2\Gamma$, and one is paid theta exactly in compensation for being short gamma.

Example 8.14

In the numerical example above ($s = K = 100$, $r = 0.05$, $\sigma = 0.2$, $\tau = 1$) we have $\varphi(d_1) = \varphi(0.35) = 0.3752$, so

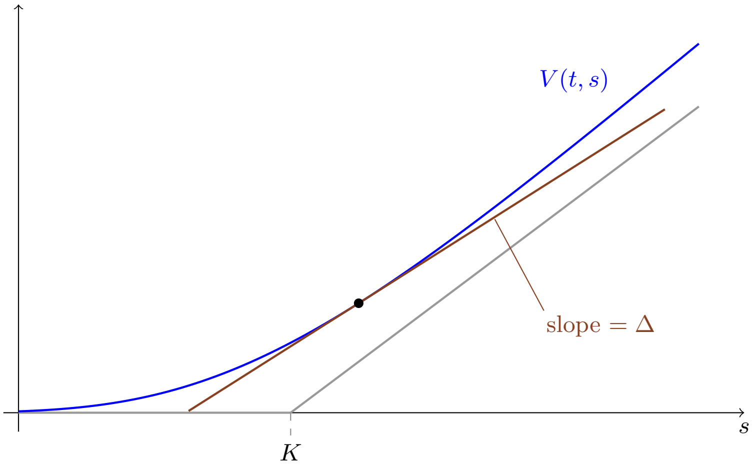
$$\Gamma = \frac{0.3752}{100 \cdot 0.2} = 0.01876, \qquad \mathcal{V} = 100 \cdot 0.3752 = 37.52,$$

$$\Theta = -\tfrac{1}{2}(100)(0.2)(0.3752) - 0.05 \cdot 95.12 \cdot 0.5596 = -6.41.$$

Checking the relation above: $-6.41 + 0.05 \cdot 100 \cdot 0.6368 + \frac{1}{2}(0.04)(10^4)(0.01876) = -6.41 + 3.18 + 3.75 = 0.52 = 0.05 \cdot 10.45$. ✓

The vega tells us that an extra percentage point of volatility is worth about £0.375 on this option – a useful rule of thumb, since a one-year at-the-money option always has $\mathcal{V} \approx 0.4s$.

Geometrically, Δ is the slope of the price curve, i.e. the number of shares whose value changes at the same rate as the option:



Remark (Why hedging works)**.** Suppose a bank sells the call for $V(0, S_0)$ and continuously holds $\Delta_t = V_s(t, S_t)$ shares, financing the rest at rate r. By the proof of Theorem 8.9 the portfolio value satisfies exactly the same stochastic differential equation as $V(t, S_t)$ and starts at the same value, so the two agree for all t; at maturity the portfolio is worth exactly the payout. The bank has eliminated its risk completely, and its profit is the difference between the price it charged and $V(0, S_0)$.

In reality one rebalances at discrete times, so there is a residual hedging error. Its size is governed by Γ: the larger the gamma, the more the delta moves between rebalancing dates and the worse the approximation. Options near the money and near expiry have very large gamma, which is why they are unpleasant to hedge.

Proposition 8.15 (Monotonicity)
If g is increasing then $\Delta \geq 0$; if g is convex then $\Gamma \geq 0$.

Proof. Both are immediate from the representation

$$V(t,s) = e^{-r\tau}\,\mathbb{E}\big[g\big(se^{(r-\sigma^2/2)\tau+\sigma\sqrt{\tau}Z}\big)\big] :$$

for each fixed value of Z the integrand is an increasing (respectively convex) function of s whenever g is, and both properties survive taking expectations. □

§8.7 Implied Volatility

Of the five inputs to the Black–Scholes formula, four – s, K, τ, r – are directly observable. The volatility σ is not. But option prices *are* observable, and the formula can be inverted.

Proposition 8.16
For fixed s, K, τ, r, the Black–Scholes call price is a strictly increasing continuous function of σ, mapping $(0, \infty)$ onto $\big((s - Ke^{-r\tau})^+,\ s\big)$.

Proof. Vega is $\mathcal{V} = s\sqrt{\tau}\varphi(d_1) > 0$, giving strict monotonicity. As $\sigma \downarrow 0$ we have $d_1, d_2 \to \pm\infty$ according to the sign of $\log(s/K) + r\tau$, and the price tends to $(s - Ke^{-r\tau})^+$, the value of the forward. As $\sigma \to \infty$, $d_1 \to \infty$ and $d_2 \to -\infty$, so the price tends to s. □

Definition 8.17 (Implied volatility)

Given an observed market price C^{mkt} of a call with strike K and maturity T, the **implied volatility** $\sigma_{\mathrm{imp}}(K,T)$ is the unique σ for which the Black–Scholes formula returns C^{mkt}.

The inversion has no closed form, but the proposition guarantees that a bisection or Newton search converges, and vega – being the derivative one needs for Newton's method – is available in closed form.

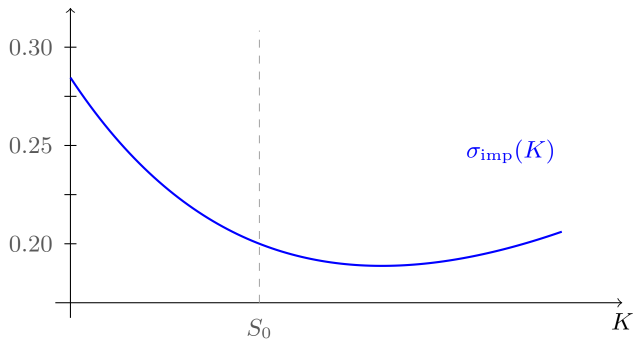
Example 8.18

Continue with $s = K = 100$, $r = 0.05$, $\tau = 1$, and suppose the option trades at $C^{\mathrm{mkt}} = 12$ rather than the 10.45 produced by $\sigma = 0.2$. One step of Newton's method from $\sigma_0 = 0.2$, using $\mathcal{V} = 37.52$ from the previous section, gives

$$\sigma_1 = 0.2 + \frac{12 - 10.45}{37.52} = 0.2413,$$

and the true implied volatility is 0.2411: a single step is already accurate to four decimal places, because the call price is very nearly linear in σ near the money. The market is therefore quoting this option as though the stock will move about 24% over the year rather than 20%.

If the Black–Scholes model were literally correct, then implied volatility would be the same for every strike and maturity, since there is only one σ in the model. It is not. Plotting σ_{imp} against strike for a fixed maturity typically gives a curve like the following, known as the **volatility smile** (or, when it is lopsided as here, the **skew**).



The market charges more than Black–Scholes for low-strike puts, because the real distribution of returns has a fatter left tail than the lognormal: crashes happen. The smile is therefore best read not as a statement about volatility but as a statement about the market's implied distribution for S_T, which is a nonparametric object recoverable from the prices of calls at all strikes. That the model is systematically wrong and yet universally used is not a paradox: implied volatility is simply a convenient reparametrisation of price, in the same way that a bond's yield is a reparametrisation of its price.

§9 Further Topics

The Black–Scholes machine is flexible, and a great deal can be done by making small modifications to it. We finish with a selection.

§9.1 Dividends

Real shares pay dividends. Suppose the stock pays a continuous dividend at rate q, meaning that holding one share over $[t, t+\,\mathrm{d}t]$ produces a cash payment $qS_t\,\mathrm{d}t$ in addition to the capital gain. Then the quantity which must be a martingale after discounting is not the price but the *total return*: reinvesting the dividends, one share at time 0 becomes e^{qt} shares at time t.

Proposition 9.1

With a continuous dividend yield q, the risk-neutral dynamics are

$$\mathrm{d}S_t = (r-q)S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}\tilde{W}_t, \qquad \text{so} \qquad S_T = S_t e^{(r-q-\sigma^2/2)\tau+\sigma(\tilde{W}_T-\tilde{W}_t)},$$

and the price of a European call is

$$C = se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_{1,2} = \frac{\log(s/K) + (r-q\pm\frac{\sigma^2}{2})\tau}{\sigma\sqrt{\tau}}.$$

Proof. The traded asset with no intermediate payments is $\hat{S}_t = e^{qt}S_t$ (the reinvested position), so $e^{-rt}\hat{S}_t$ must be a $\mathbb{Q}$-martingale, i.e. $e^{-(r-q)t}S_t$ is a $\mathbb{Q}$-martingale. This forces the stated drift. The formula then follows from Theorem 8.4 on replacing s by $se^{-q\tau}$ and r by r throughout the derivation; equivalently, on noting that $\mathbb{E}_{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = se^{(r-q)\tau}$. □

Dividends break the 'never exercise a call early' result of §6.7. The reason is transparent: an American call holder receives no dividends, so if q is large enough it can be worth exercising to capture them. Precisely, the submartingale argument required $\mathbb{E}_{\mathbb{Q}}[S_n \mid \mathcal{F}_{n-1}] \geq (1+r)S_{n-1}$, which fails when $q > 0$.

§9.2 Foreign Exchange

Consider two currencies, 'domestic' and 'foreign', with riskless rates r_d and r_f. Let X_t be the exchange rate, quoted as the number of units of domestic currency per unit of foreign currency.

A domestic investor may hold the foreign bank account. In domestic terms it is worth $X_t e^{r_f t}$, and it is a traded asset with no intermediate payments. Hence, under the domestic risk-neutral measure $\mathbb{Q}_d$, the process $e^{-r_d t}X_t e^{r_f t}$ must be a martingale. So X behaves exactly like a stock paying a dividend at rate r_f:

$$\mathrm{d}X_t = (r_d - r_f)X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}\tilde{W}_t.$$

Proposition 9.2 (Garman–Kohlhagen)

The domestic price of a European call on the exchange rate with strike K and maturity T is

$$C = xe^{-r_f\tau}\Phi(d_1) - Ke^{-r_d\tau}\Phi(d_2), \qquad d_{1,2} = \frac{\log(x/K) + (r_d - r_f \pm \frac{\sigma^2}{2})\tau}{\sigma\sqrt{\tau}}.$$

There is a pleasant symmetry here which is worth spelling out, because it is a genuine change of numeraire. The foreign investor sees the exchange rate $Y_t = 1/X_t$, and by Itô

$$dY_t = (r_f - r_d + \sigma^2)Y_t\, dt - \sigma Y_t\, d\tilde{W}_t$$

under $\mathbb{Q}_d$. The extra σ^2 looks alarming – it appears to say that the two investors disagree about which currency is expected to appreciate, a puzzle known as **Siegel's paradox**. The resolution is that the foreign investor prices with a different measure $\mathbb{Q}_f$, defined by

$$\frac{d\mathbb{Q}_f}{d\mathbb{Q}_d} = \frac{e^{-r_f t} X_t^{-1}}{X_0^{-1} e^{-r_d t}},$$

the density being (the discounted value of) one unit of domestic currency expressed in foreign terms. Under $\mathbb{Q}_f$ one finds $dY_t = (r_f - r_d)Y_t\, dt + \sigma Y_t\, d\hat{W}_t$ for a $\mathbb{Q}_f$-Brownian motion $\hat{W}$, restoring the symmetry. There is no paradox: expectations are measure-dependent, and prices are not.

§9.3 American Options in Continuous Time

In continuous time the price of an American claim with payout $g(S_t)$ is

$$\pi_t = \sup_{t \le \mathcal{T} \le T} \mathbb{E}_{\mathbb{Q}}\left[e^{-r(\mathcal{T}-t)} g(S_{\mathcal{T}}) \mid \mathcal{F}_t\right],$$

the supremum over stopping times. As in discrete time, the value process is the smallest supermartingale dominating the discounted payout. The analytic formulation is a **free boundary problem**: there is a **continuation region** $\mathcal{C}$ in which V solves the Black–Scholes PDE and $V > g$, and a **stopping region** in which $V = g$; the boundary between them is not known in advance and must be found as part of the solution.

At the boundary one imposes **value matching** $V = g$ and **smooth fit** $V' = g'$. The smooth fit condition is not a boundary condition in the usual sense – it is an optimality condition, and it is what selects the correct boundary from the one-parameter family of candidates. The case which can be solved completely is the following.

Example 9.3 (The perpetual American put)

Take $g(s) = (K - s)^+$ and no expiry date. By stationarity the value $V(s)$ does not depend on t, and the exercise rule must be of the form 'exercise when $S_t \le b$' for some level b to be determined. In the continuation region $s > b$, the Black–Scholes PDE with $V_t = 0$ becomes the Euler equation

$$rsV'(s) + \tfrac{1}{2}\sigma^2 s^2 V''(s) = rV(s).$$

Trying $V(s) = s^{-\gamma}$ gives the indicial equation $\frac{1}{2}\sigma^2\gamma(\gamma+1) - r\gamma - r = 0$, whose positive root is

$$\gamma = \frac{2r}{\sigma^2}.$$

(The other root is $\gamma = -1$, giving the solution $V = s$, which is excluded because V must be bounded as $s \to \infty$.) So $V(s) = As^{-\gamma}$ for $s > b$. Value matching at $s = b$ gives $Ab^{-\gamma} = K - b$, i.e.

$$V(s) = (K - b)\left(\frac{b}{s}\right)^{\gamma}, \qquad s \ge b.$$

Smooth fit requires $V'(b) = -1$, i.e. $-\gamma(K-b)/b = -1$, giving

$$b = \frac{\gamma K}{1+\gamma} = \frac{2rK}{2r+\sigma^2}.$$

Note the sensible limits: as $\sigma \to 0$ we get $b \to K$ (with no volatility there is no reason to wait), and as $\sigma \to \infty$ we get $b \to 0$ (with enormous volatility it is always worth waiting). For finite expiry no closed form exists and one must compute numerically.

§9.4 Barrier and Path-Dependent Claims

Not every claim is a function of S_T alone. A **barrier option** pays off only if the price has (or has not) crossed a level B before maturity: the **up-and-in** version of a claim Y pays $Y\mathbb{1}_{\{\max_{t\le T} S_t \ge B\}}$, the **up-and-out** version pays $Y\mathbb{1}_{\{\max_{t\le T} S_t < B\}}$, and similarly for down-and-in and down-and-out. Adding an in and an out version recovers the original claim, so it suffices to price one of them.

Since $S_t = S_0 e^{\sigma(\tilde{W}_t + ct)}$ with $c = r/\sigma - \sigma/2$ under $\mathbb{Q}$, barrier crossing is a statement about the maximum of a Brownian motion with drift, and the tools of §7 apply.

Proposition 9.4

In the Black–Scholes model, the initial price of the up-and-out claim with payout $g(S_T)\mathbb{1}_{\{\max_{t\le T} S_t < B\}}$ equals the price of the European claim with payout

$$g(S_T)\mathbb{1}_{\{S_T<B\}} - \left(\frac{B}{S_0}\right)^{2r/\sigma^2-1} g\left(\frac{B^2 S_T}{S_0^2}\right)\mathbb{1}_{\{S_T<S_0^2/B\}}.$$

Proof sketch. Write $b = \log(B/S_0)/\sigma$ and $S_t = S_0 e^{\sigma(W_t+ct)}$ for a $\mathbb{Q}$-Brownian motion W. Since $\{\max_t S_t < B\} \subseteq \{S_T < B\}$,

$$g(S_T)\mathbb{1}_{\{\max_t S_t<B\}} = g(S_T)\mathbb{1}_{\{S_T<B\}} - g(S_T)\mathbb{1}_{\{S_T<B,\ \max_t S_t\ge B\}},$$

so it suffices to compute the expectation of the last term. By the Cameron–Martin theorem (Theorem 7.20) we may strip off the drift, then apply the reflection lemma (Lemma 7.17) to the driftless maximum, then reinstate the drift:

$$\begin{aligned}
\mathbb{E}\big[g(S_T)\mathbb{1}_{\{S_T<B,\max_t S_t\ge B\}}\big] &= \mathbb{E}\big[e^{cW_T - c^2T/2} g(S_0 e^{\sigma W_T})\mathbb{1}_{\{W_T<b,\ \max_t W_t \ge b\}}\big] \\
&= \mathbb{E}\big[e^{c(2b-W_T)-c^2T/2} g(S_0 e^{\sigma(2b-W_T)})\mathbb{1}_{\{W_T>b\}}\big] \\
&= e^{2bc}\,\mathbb{E}\big[e^{-cW_T - c^2T/2} g(S_0 e^{2b\sigma} e^{-\sigma W_T})\mathbb{1}_{\{W_T>b\}}\big],
\end{aligned}$$

and undoing the change of measure gives the stated European payout, with $e^{2bc} = (B/S_0)^{2r/\sigma^2-1}$ and $S_0 e^{2b\sigma} = B^2/S_0$. □

Example 9.5 (Forward-start call)

A **forward-start** call fixes its strike to be the price at an intermediate date $T_0 < T_1$, so the payout is $(S_{T_1} - S_{T_0})^+$. For $t \ge T_0$ this is an ordinary call with known strike. For $t < T_0$, using that S_{T_1}/S_{T_0} is independent of $\mathcal{F}_{T_0}$ under $\mathbb{Q}$ (the increments of $\tilde{W}$

being independent),

$$\begin{aligned}\pi_t &= e^{-r(T_1-t)}\mathbb{E}_{\mathbb{Q}}\left[(S_{T_1}-S_{T_0})^+ \mid \mathcal{F}_t\right]\\ &= e^{-r(T_1-t)}\mathbb{E}_{\mathbb{Q}}\left[S_{T_0}\left(\frac{S_{T_1}}{S_{T_0}}-1\right)^+ \;\Big|\; \mathcal{F}_t\right]\\ &= e^{-r(T_1-t)}\,\mathbb{E}_{\mathbb{Q}}[S_{T_0}\mid\mathcal{F}_t]\,\mathbb{E}_{\mathbb{Q}}\left[\left(\frac{S_{T_1}}{S_{T_0}}-1\right)^+\right]\\ &= e^{-r(T_1-T_0)}S_t\,\mathbb{E}_{\mathbb{Q}}\left[\left(\frac{S_{T_1}}{S_{T_0}}-1\right)^+\right],\end{aligned}$$

and the remaining expectation is the Black–Scholes price of an at-the-money call on a unit stock over the horizon T_1-T_0. So the forward-start call is priced by a homogeneity argument, with no new integration required.

§9.5 Numerical Schemes

Beyond a small collection of examples, option prices must be computed numerically. There are three standard approaches: Monte Carlo simulation of the risk-neutral expectation; the binomial tree of §6, which is simply an explicit finite-difference scheme; and direct solution of the Black–Scholes PDE. We say a word about the last.

After the change of variables reducing Black–Scholes to the heat equation $u_t=\frac{1}{2}u_{xx}$, one discretises on a grid with steps Δt and Δx, writing $U_{n,k}\approx u(n\Delta t,k\Delta x)$. The three standard discretisations are

$$\begin{aligned}&\text{(forward)} && \frac{U_{n+1,k}-U_{n,k}}{\Delta t}=\frac{U_{n,k+1}-2U_{n,k}+U_{n,k-1}}{2(\Delta x)^2},\\ &\text{(backward)} && \frac{U_{n+1,k}-U_{n,k}}{\Delta t}=\frac{U_{n+1,k+1}-2U_{n+1,k}+U_{n+1,k-1}}{2(\Delta x)^2},\\ &\text{(Crank–Nicolson)} && \frac{U_{n+1,k}-U_{n,k}}{\Delta t}=\frac{U_{n,k+1}-2U_{n,k}+U_{n,k-1}}{4(\Delta x)^2}\\ &&&\qquad\qquad+\frac{U_{n+1,k+1}-2U_{n+1,k}+U_{n+1,k-1}}{4(\Delta x)^2},\end{aligned}$$

the last being the average of the first two.

Definition 9.6 (von Neumann stability)

A scheme is **stable** in the von Neumann sense if $\sup_\beta|\varrho(\beta)|\le 1$, where $\varrho(\beta)$ is defined by requiring that $U_{n,k}=\varrho(\beta)^n e^{i\beta k\Delta x}$ solves the recursion.

Substituting the ansatz and writing $\nu=\Delta t/(\Delta x)^2$ gives

$$\varrho_{\mathrm{fwd}}(\beta)=1-\nu\big(1-\cos(\beta\Delta x)\big),\qquad \varrho_{\mathrm{bwd}}(\beta)=\frac{1}{1+\nu(1-\cos(\beta\Delta x))},$$

$$\varrho_{\mathrm{CN}}(\beta)=\frac{2-\nu(1-\cos(\beta\Delta x))}{2+\nu(1-\cos(\beta\Delta x))}.$$

The forward scheme is stable only if $\nu\le 1$, which forces very small time steps; the backward and Crank–Nicolson schemes are unconditionally stable, at the cost of solving a tridiagonal linear system at each step. Crank–Nicolson is additionally second-order accurate in time, and is the usual default.

Remark. It is a pleasing coda that the binomial model, which we introduced as a self-contained discrete market, reappears here as a numerical method for the continuous one; and that the stability condition $\nu \leq 1$ is exactly the requirement that the implied risk-neutral probabilities lie in $[0, 1]$, i.e. that the discretised model be arbitrage-free. Numerical stability and the absence of arbitrage are, in this instance, the same condition.